HEAVY HAUL SEMINAR • APRIL 23-24, 2026

Data-Driven Assessment of Rail Grinding Frequency for Sustainable Track Integrity in India's Dedicated Freight Corridor

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DALLAS, TX

Dedicated Freight Corridor

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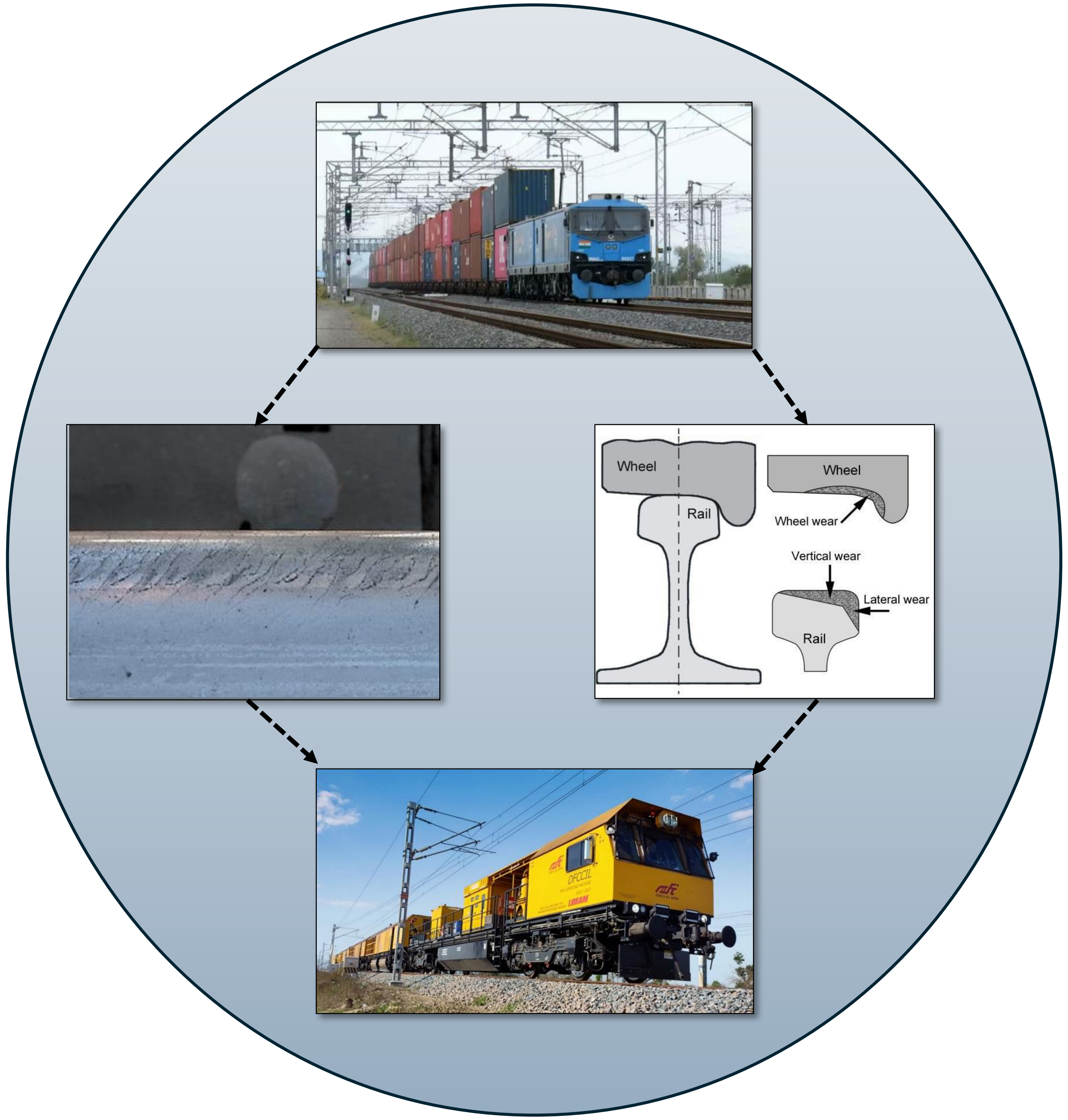
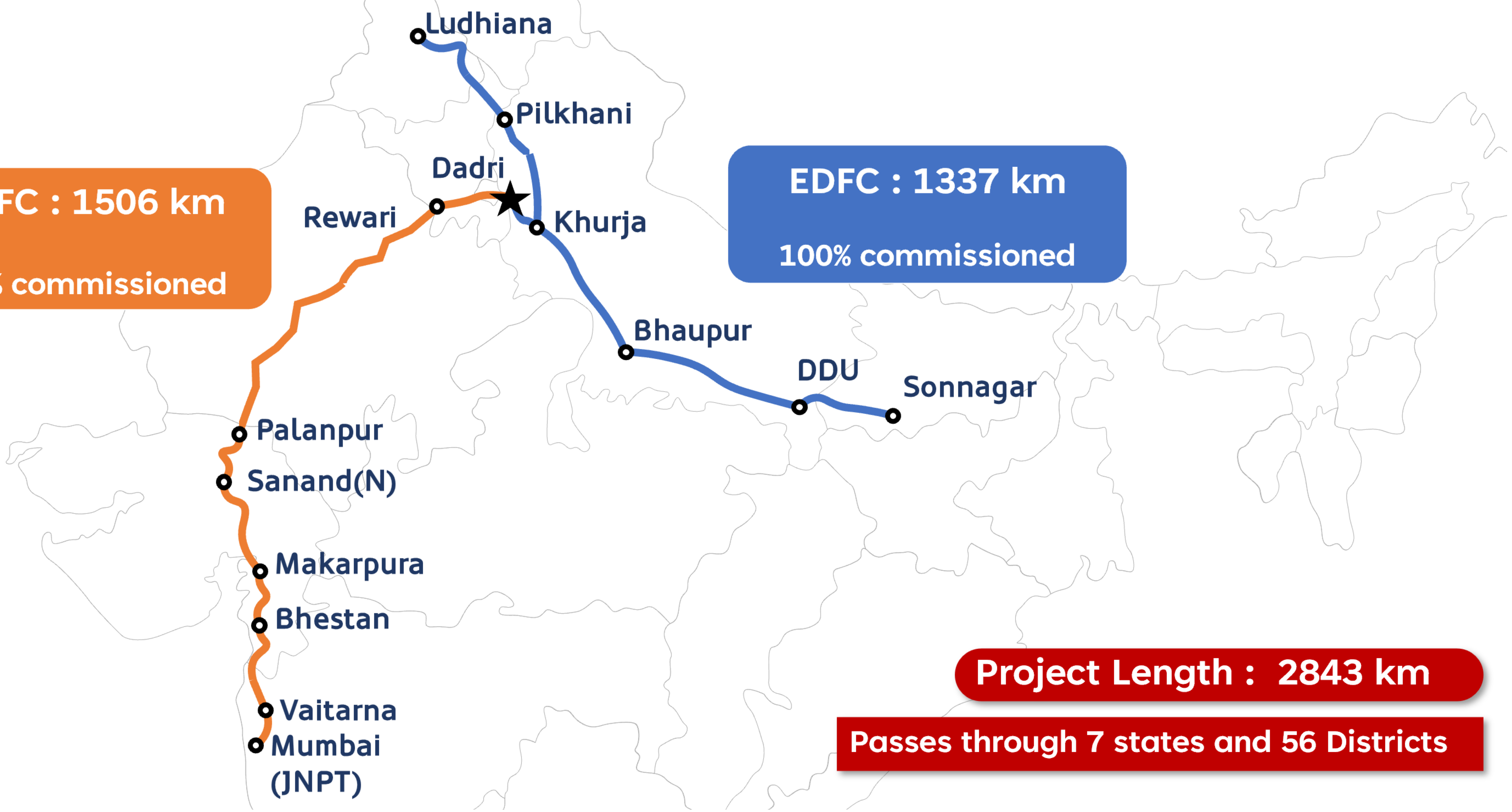
Contents



DFC Project Overview



WDFC : 1506 km

100% commissioned

EDFC : 1337 km

100% commissioned

Project Length : 2843 km

Passes through 7 states and 56 Districts

Dedicated Freight Corridor

Introduction

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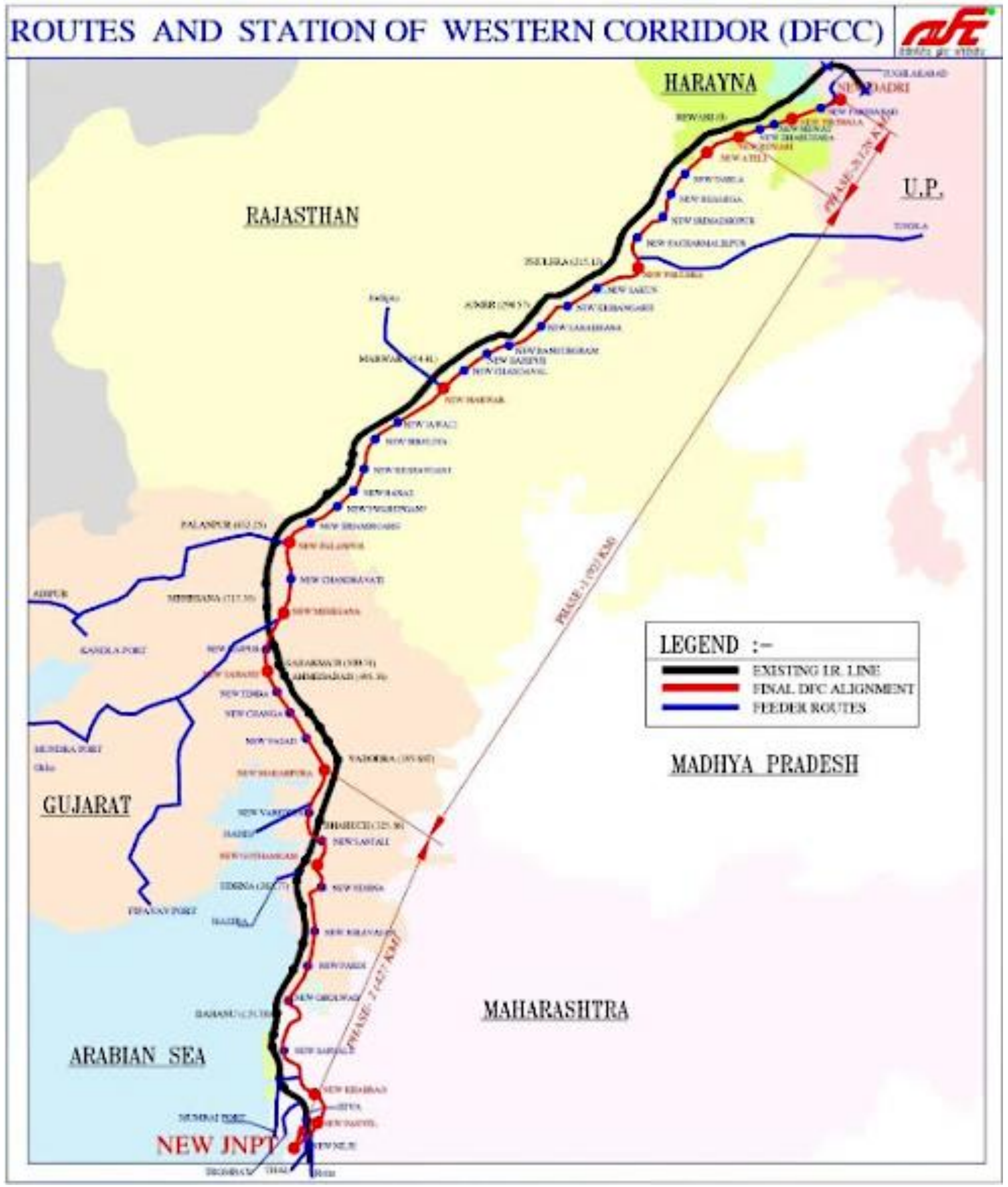

*The **Western Dedicated Freight Corridor (WDFC)** is one of the two flagship freight railway corridors developed by the Dedicated Freight Corridor Corporation of India Ltd. (DFCCIL) under the Ministry of Railways.*

Divisional Structure of WDFC:

1. **Mumbai Division (JNPT to Vaitarna) – Maharashtra**
2. **Vadodara Division – Gujarat**
3. **Ahmedabad Division – Gujarat**
4. **Ajmer Division – Rajasthan**
5. **Jaipur Division – Rajasthan**
6. **Rewari Division – Haryana**
7. **Dadri Section (UP) – Uttar Pradesh**

Jawaharlal Nehru Port Terminal (JNPT),
Navi Mumbai (Maharashtra) to Dadri (Uttar Pradesh)
1506 Kilometres

The observations must be viewed in the context of rapidly increasing traffic and evolving maintenance deployment strategies.

Currently, rail grinding operations are being managed with a single 72-stone RGM, and capacity augmentation is underway to align with network-wide requirements

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Introduction Contd.

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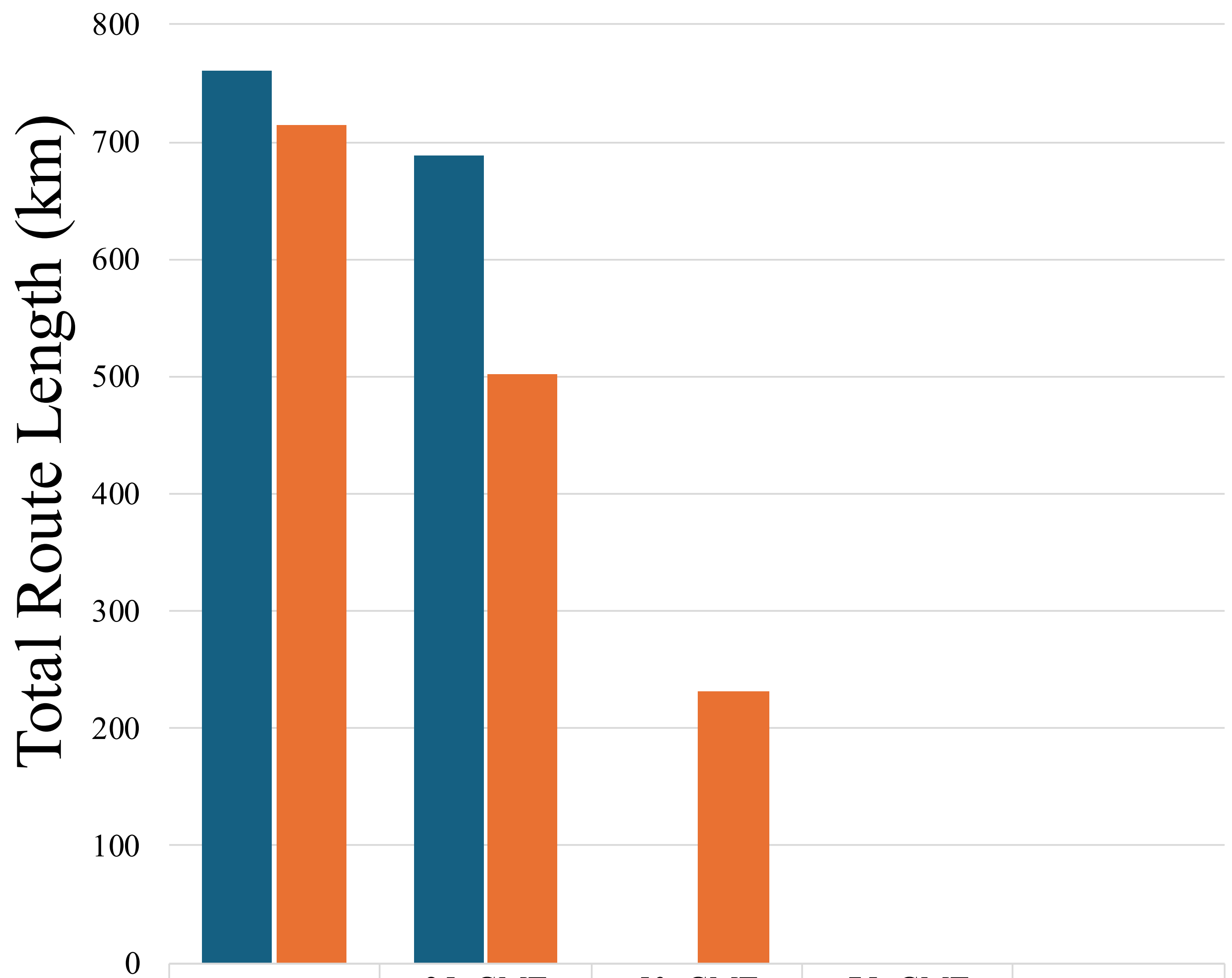
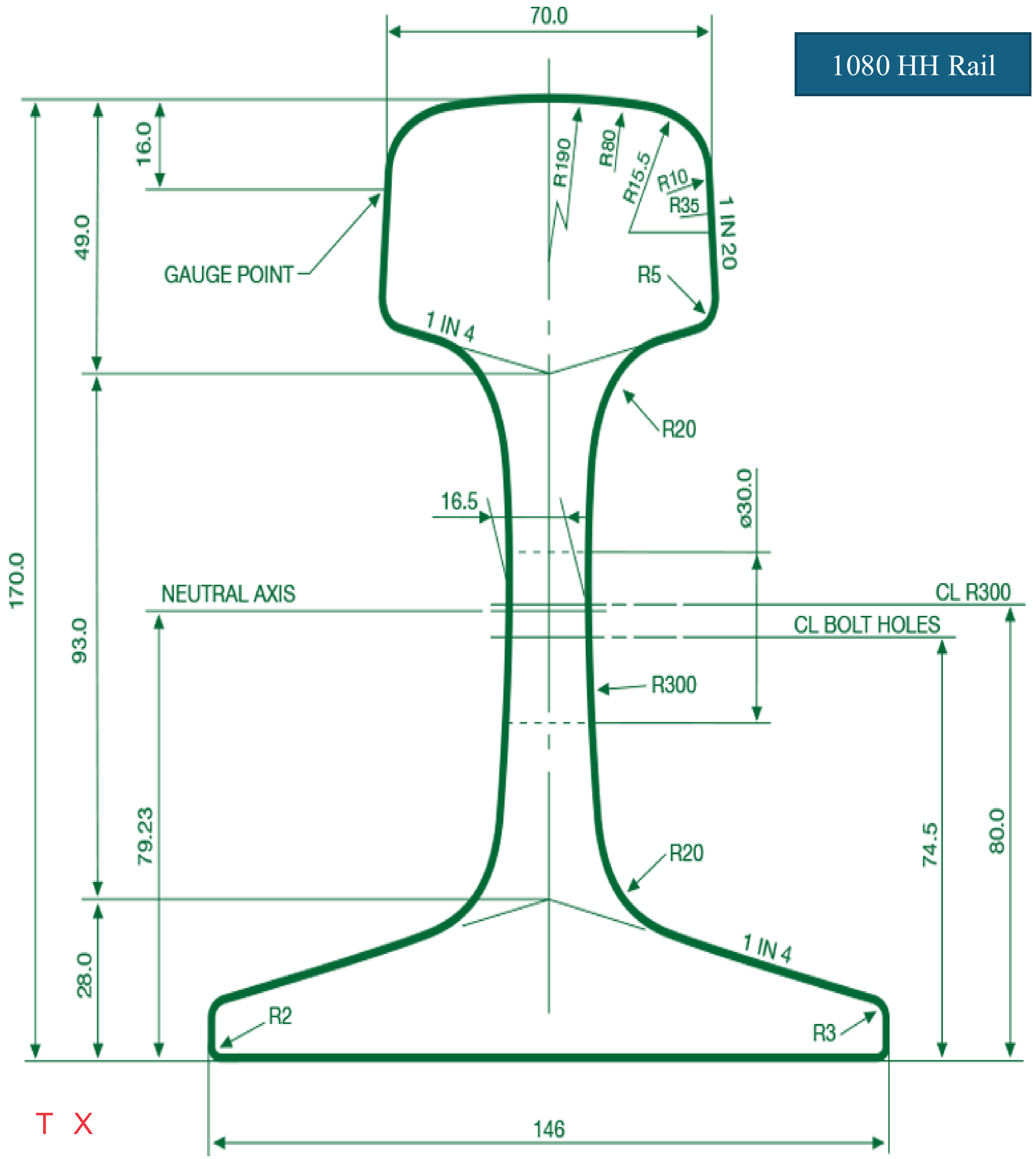
WDFC



	<25	25≤GMT <50	50≤GMT <75	75≤GMT <100	100≤
Total Section Length UP	760.645	688.0425	0	0	0
Total Section Length DN	715.001	502.5314	231.1571	0	0

For WDFC DN line 502 km and 231 km of track length observes annual GMT between **25-50 GMT** and between **50-75 GMT** respectively.

1080 HH Rail



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Introduction Contd.

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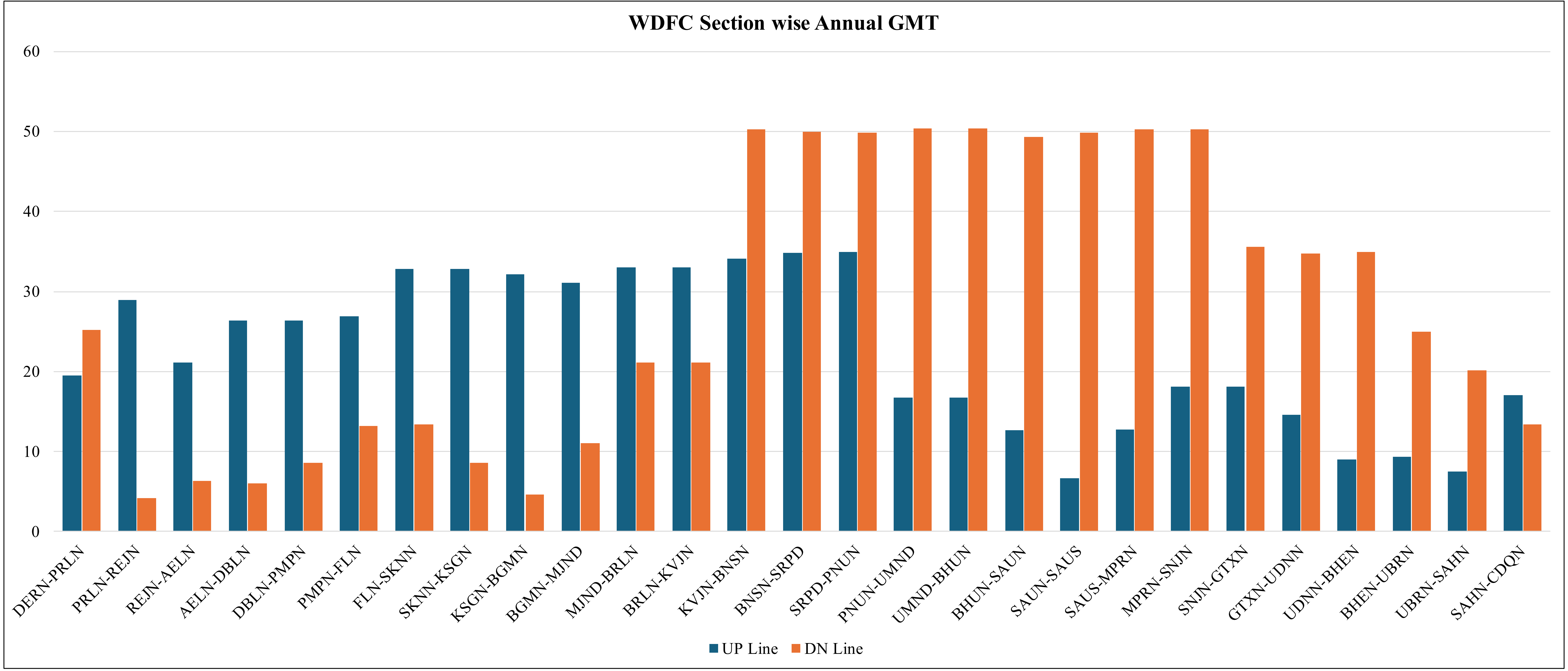

→For UP line annual GMT varies from **6.67 GMT** to **34.91 GMT** for **SAUN-SAUS** and **SPRD-PNUN** sections respectively.

→For DN line annual GMT varies from **4.16 GMT** to **50.40 GMT** for **PRLN-REJN** and **PNUN-UMND** sections respectively

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Introduction Contd.

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- DFCCIL has outsourced complete operation and maintenance of the Rail Grinding Machines to a specialist O&M Service Provider – M/s Vandhana International Pvt. Ltd.

- They will be responsible for managing the O&M for the next 12 years along with the grinding machine OEM (M/s Loram Maintenance of Way, Inc. USA)

- Due to lack of availability of current assets in our fleet, we needed to determine the best grinding cycle frequency and we wanted to rely on our actual data to determine the next steps and making the grinding program more scientific.

- The paper is a reflection of some aspects of the work which are concluded and other aspects which are still under development – including aspects linked to Grind Quality Index during pre inspection, Surface Scoring, Defect Generation Rate with Ultrasonic testing etc.

Methodology

Data	Source
GMT Data	DFCCIL Operations Team
Rail Profile Data	MiniProf (MP – 262) used for data collection for high accuracy and repeatability.
DPT	SPOTCHECK DPT Kits

MP – 262

DPT Kit

Data Collection details:

Location	August 2025	October 2025	January 2026
AJMER	✓	✓	✓
JAIPUR	✓	✓	✓

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Methodology Contd.

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Wear rate measurement:

$$Wear\ rate_{(Before\ Grinding)} = \frac{Wear\ before\ GC1\ - Wear\ at\ 1st\ Collection}{GMT\ Passed} x100$$

$$Wear\ rate_{(After\ Grinding)} = \frac{Wear\ at\ 2nd\ Collection\ - Wear\ at\ 1st\ Collection}{GMT\ Passed} x100$$

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Evolution of GMT and deferred Grind cycles

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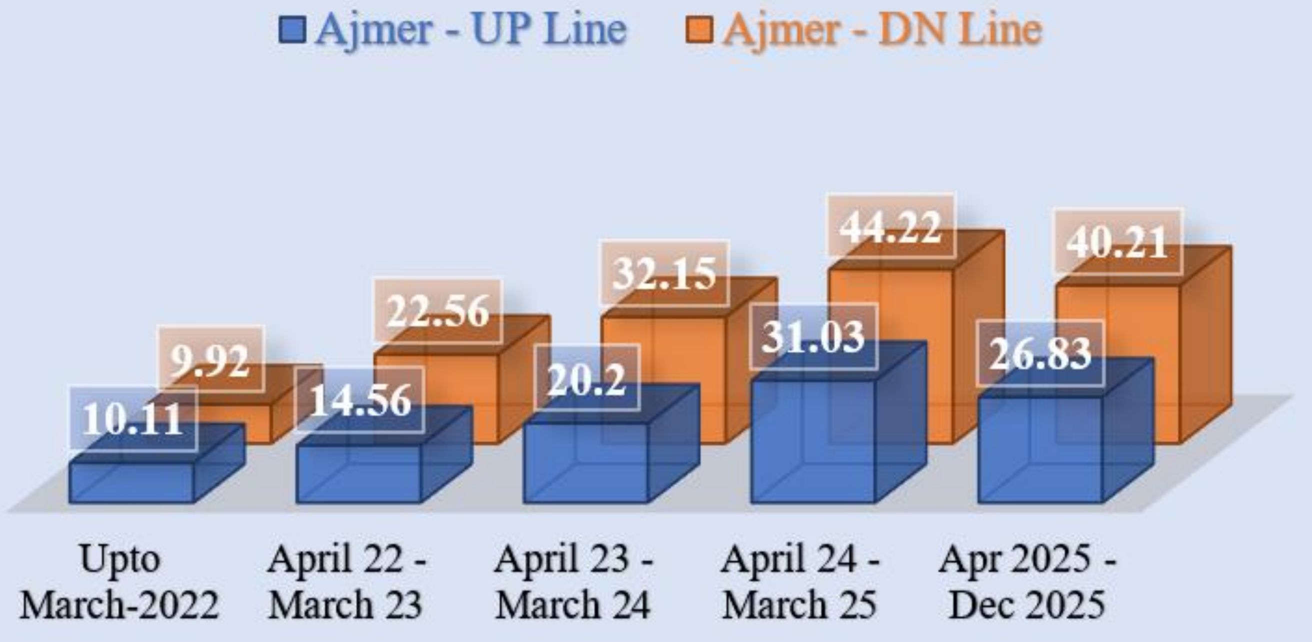
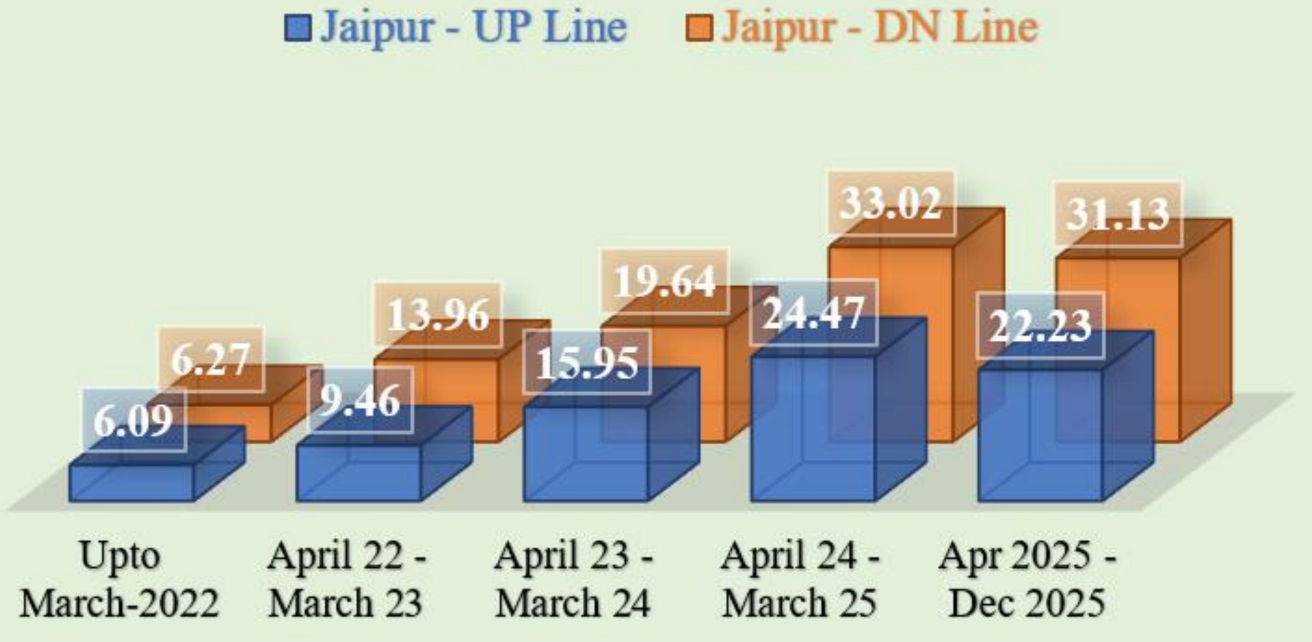
Evolution of Traffic (GMT) – AJMER Div.



Evolution of Traffic (GMT) – JAIPUR Div.



Div.	Grinding Date 1st Cycle	Avg. GMT after Grinding	Avg. Cumulative GMT	Potential Grinding Cycles Overdue
Jaipur (UP)	May - 24	43.39	79.08	1 Grind Cycle Deferred (After Initial Single Pass Grinding)
Jaipur (DN)	May - 24	58.65	104.96	2 Grind Cycle Deferred (After Initial Single Pass Grinding)
Ajmer (UP)	Mar - 24	59.01	103.95	2 Grind Cycle Deferred (After Initial Single Pass Grinding)
Ajmer (DN)	Mar - 24	88.01	150.51	3 Grind Cycle Deferred (After Initial Single Pass Grinding)

- A steady increase in traffic is observed in both divisions from the initial operational phase up to Dec 2025. For the purpose of this table, GMT has been calculated up to December 2025.
- We have assumed 25 GMT cycles for the initial 3 cycles in line with RDSO recommendations.
- In Jaipur, the UP line accumulated an average GMT of 43.39 (cumulative 79.08), leading to deferment of one grinding cycle, while the DN line recorded a higher average GMT of 58.65 (cumulative 104.96), resulting in deferment of two cycles.
- In Ajmer, the UP line reached an average GMT of 59.01 (cumulative 103.95), skipping two cycles, whereas the DN line recorded the highest loading with an average GMT of 88.01 (cumulative 150.51), corresponding to three skipped grinding cycles.

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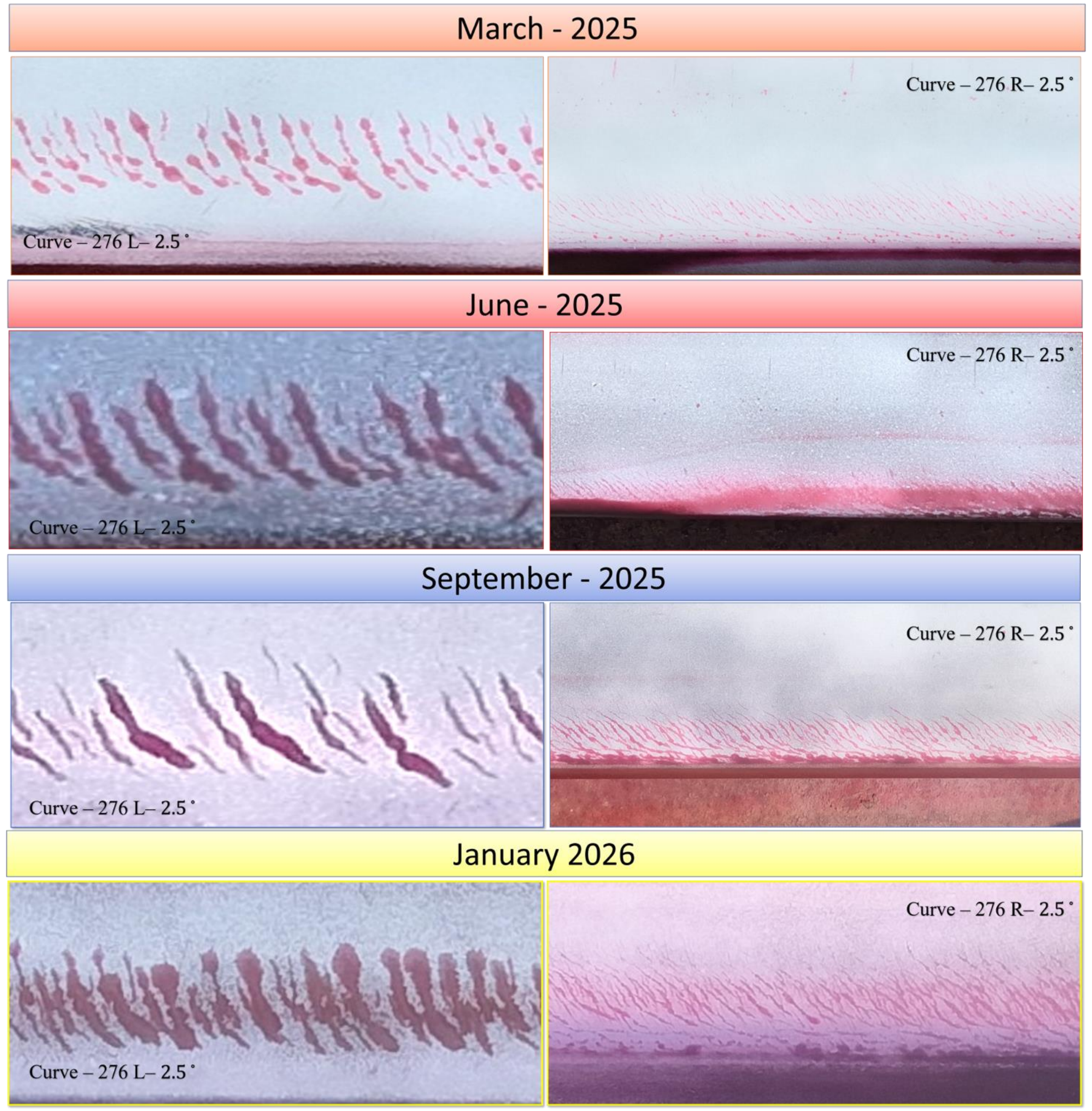
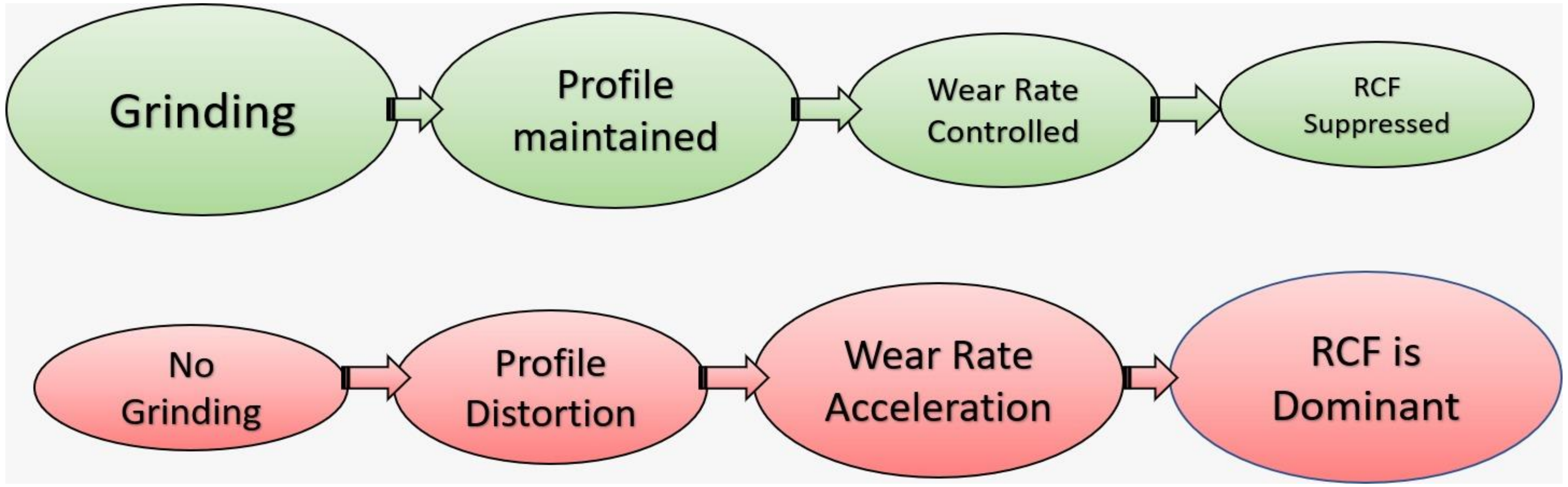
Field validated RCF Lifecycle

First Grind Cycle (Mar 2024)

Start of Grind Cycle Skip date (Sep 24)





- Immediately after grinding, the rail surfaces appear smooth with negligible damage
- By March 2025, early signs of rolling contact fatigue (RCF) emerge
- By June 2025, reflecting active propagation under cyclic loading
- By September 2025, the crack networks become deeper and more pronounced, indicating subsurface fatigue progression and an increased risk of spalling
- By January 2026, both rails exhibit significant deterioration as RCF becomes denser and visually longer
- The figure illustrates the contrasting degradation pathways under timely grinding and grinding deferral, highlighting a fundamental shift in the failure mechanism.
- This shift has significant maintenance implications, as RCF damage is more difficult to arrest once established

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Field validated RCF Lifecycle Contd.

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Stage	Time Range	Condition of RCF	GMT Range
Post Grinding	Mar 2024 – Sep 2024	Clean surface and Profile	0 – 22
Initiation of Grinding deferral	Sep 2024 – Mar 2025	Initiation of RCF	22 – 44
9 months after Grinding deferral	March 2025 – June 2025	Moderate Head checks	44 – 60
12 months after Grinding deferral	June 2025 – September 2025	Expansion in defect dimensions	60 – 74
16 months after grinding deferral	September 2025 – Jan 2026	Increase in density and frequency	74 – 88
20 months after grinding deferral	Jan 2026 – May 2026	Significant Spalling and Shelling	100+
24 months after grinding deferral	May 2026 – Sep 2026	Potential Risk of Failure due to RCF related defects	120+

- This progression can be broadly classified into four stages
- Post-grinding, characterized by a clean and stable surface within approximately 0-22 GMT
- Initiation, marked by the appearance of head checks as traffic accumulates to around 22-44 GMT
- Propagation, where crack density and severity increase significantly in the range of approximately 44-88 GMT
- Failure risks significantly increase associated with significant shelling and spalling beyond 100 GMT

Damage acceleration Behaviour analysis

Three sharp curves were selected for the damage acceleration behaviour analysis (details below):

S. No	Corridor	Segment	Curve No.	Degree	KM From	KM to
1.	WDFC	Sharp	207 L	2.5°	962.541	963.197
2.	WDFC	Sharp	280 R	2.5°	1047.01	1047.39
3.	WDFC	Sharp	276 L	2.5°	1043.724	1044.135

Data Collection details:

S. No	Data utilized	Condition	Date
1.	Rail Profile	Post Grind Cycle 1 – 22 GMT has passed	All sites in Ajmer DN Line where 88 GMT has passed since grinding. Data from Aug 2024 till Jan 2026
2.	Rail Profile	Post Grind Cycle 1 – 44 GMT has passed	
3.	Rail Profile	Post Grind Cycle 1 – 60 GMT has passed	
4.	Rail Profile	Post Grind Cycle 1 – 74 GMT has passed	
5.	Rail Profile	Post Grind Cycle 1 – 88 GMT has passed	

Rail profiles collected at different times → Wear rates calculated per 100 GMT → Inflection point determined from wear rate data

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Wear Values

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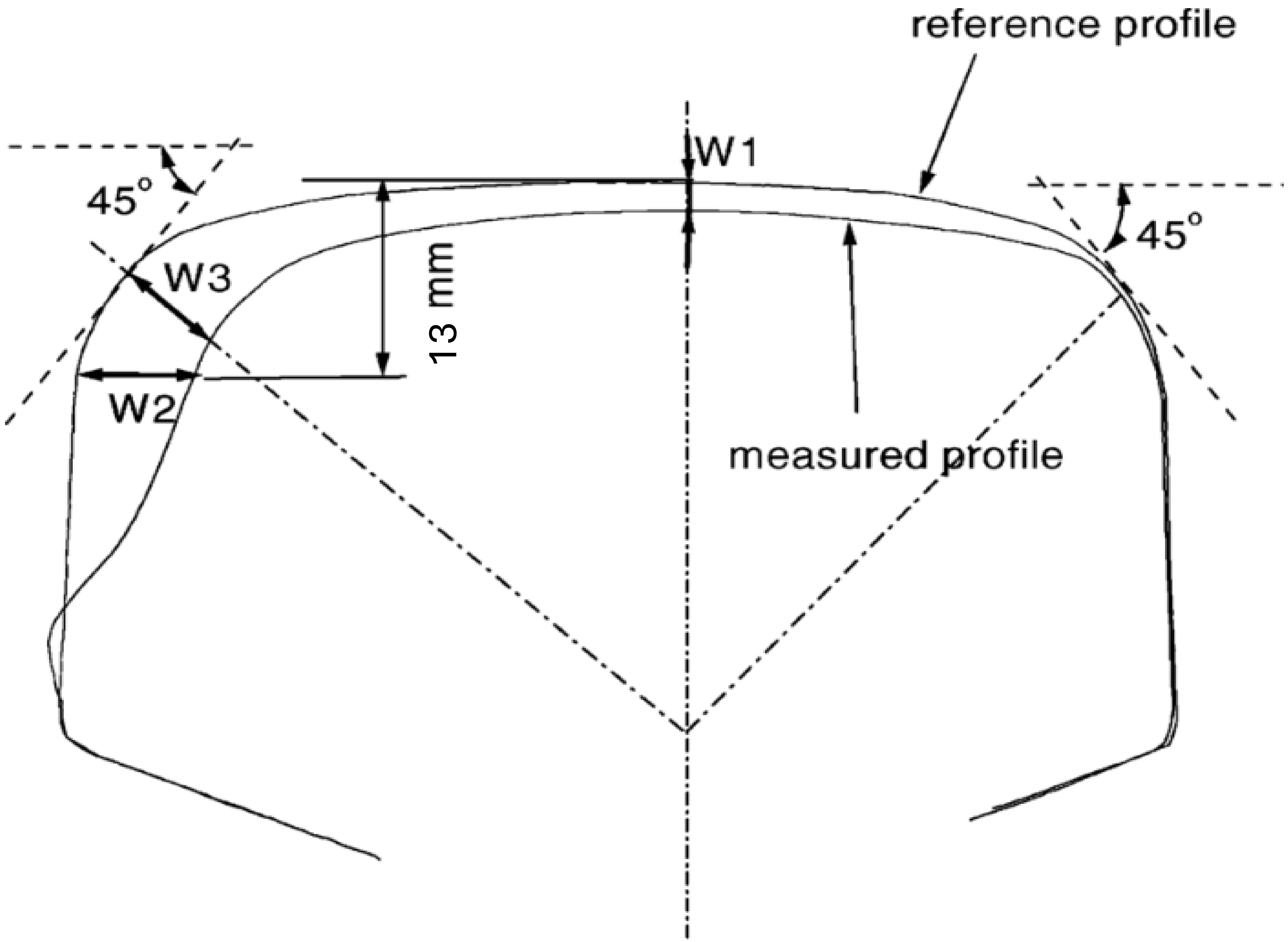

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Damage acceleration Behaviour Contd.

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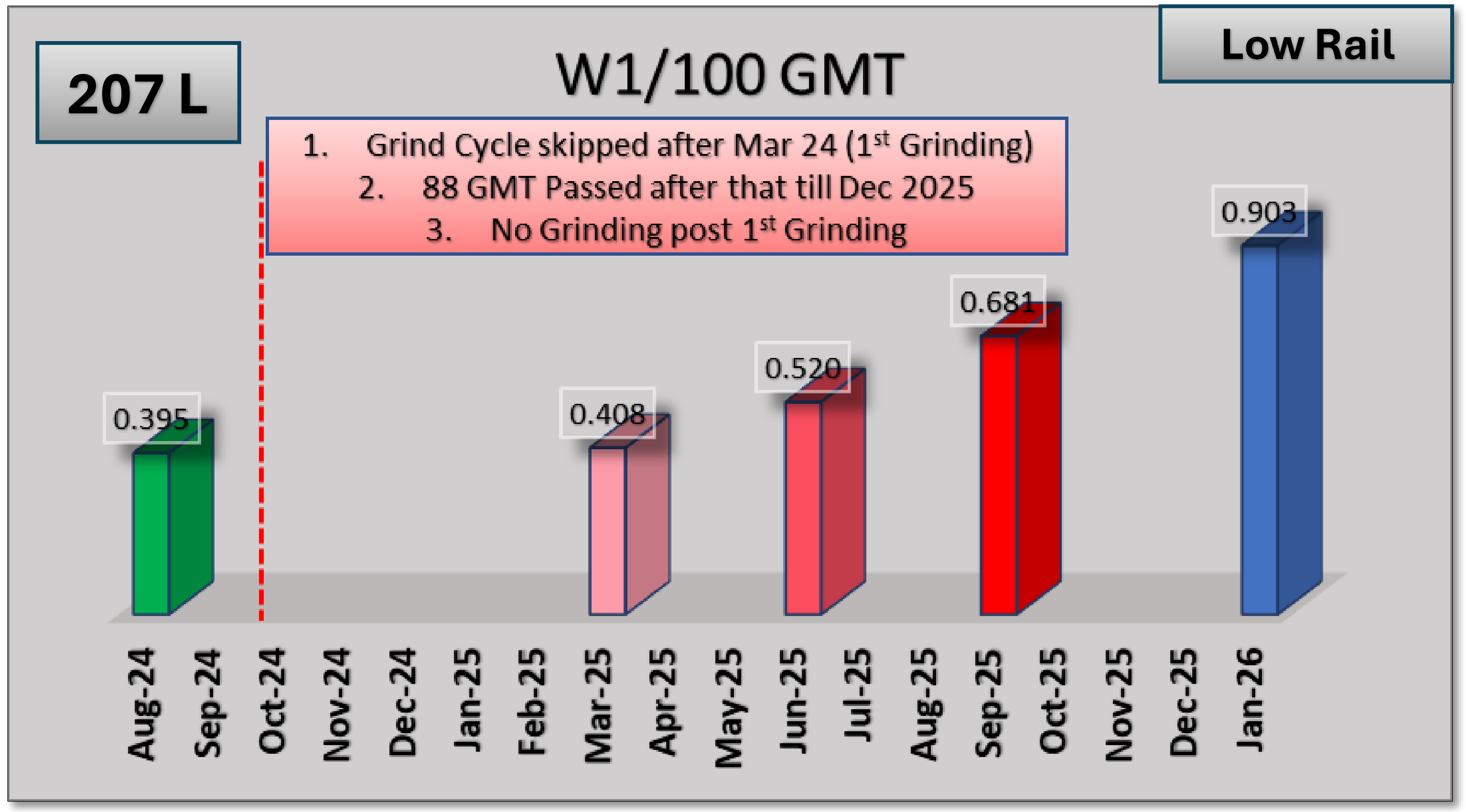

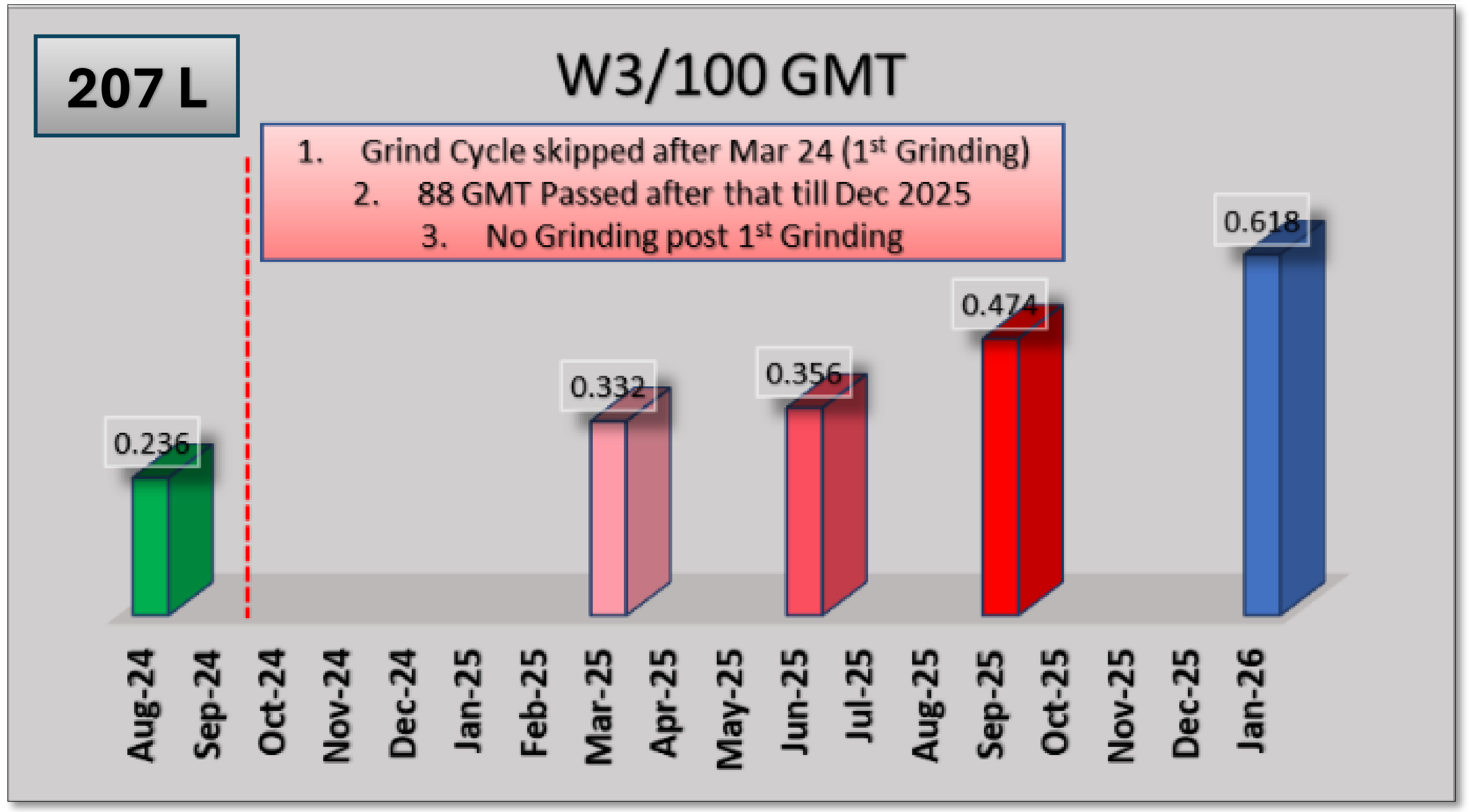

- The data indicate a consistent increase in wear rate with the accumulation of GMT.
- For W1, the low rail wear rate increased progressively from 0.395 in Aug-24 to 0.903 by Jan-26
- A similar trend is observed for W3, where the low rail wear rate increased from 0.236 in Aug-24 to 0.618 by Jan-26.

Increase in wear rate (%) – Curve 207L

Test Site	Rail head angle	Mar 25 – Jun 25	Jun 25 – Sep 25	Sep 25 – Jan 26
207L - Low	0˚ (W1)	27.5	30.9	32.63
207L - Low	45˚ (W3)	7.2	33.3	30.33

- For the low rail, the wear rate increased progressively under W1 (0°) from 27.5% (Mar–Jun 2025) to 30.9% (Jun–Sep 2025) and 32.63% (Sep 2025–Jan 2026)
- while under W3 (45°) it increased from 7.2% to 33.3% and 30.33%

Dedicated Freight Corridor

Damage acceleration Behaviour Contd.

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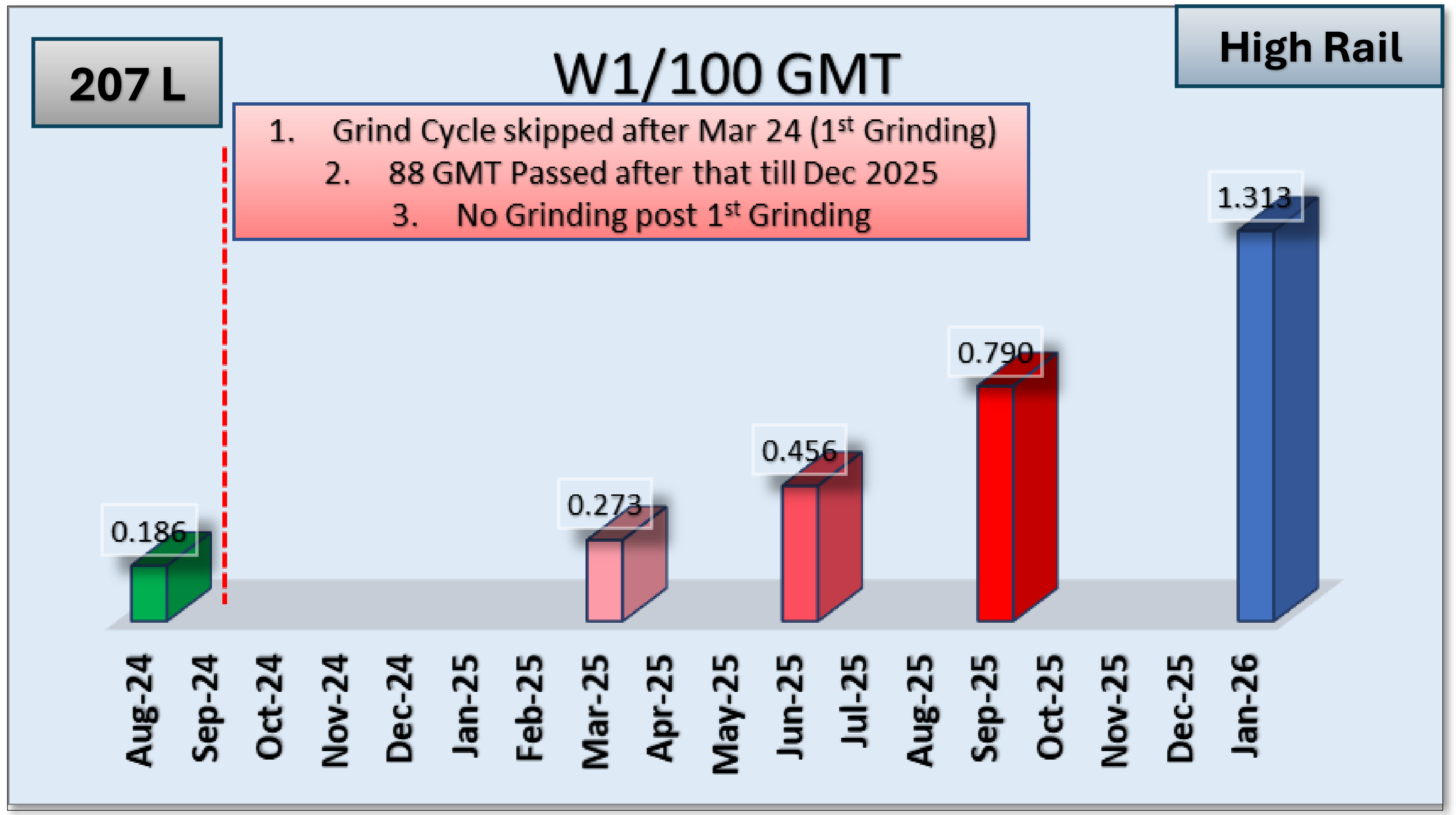

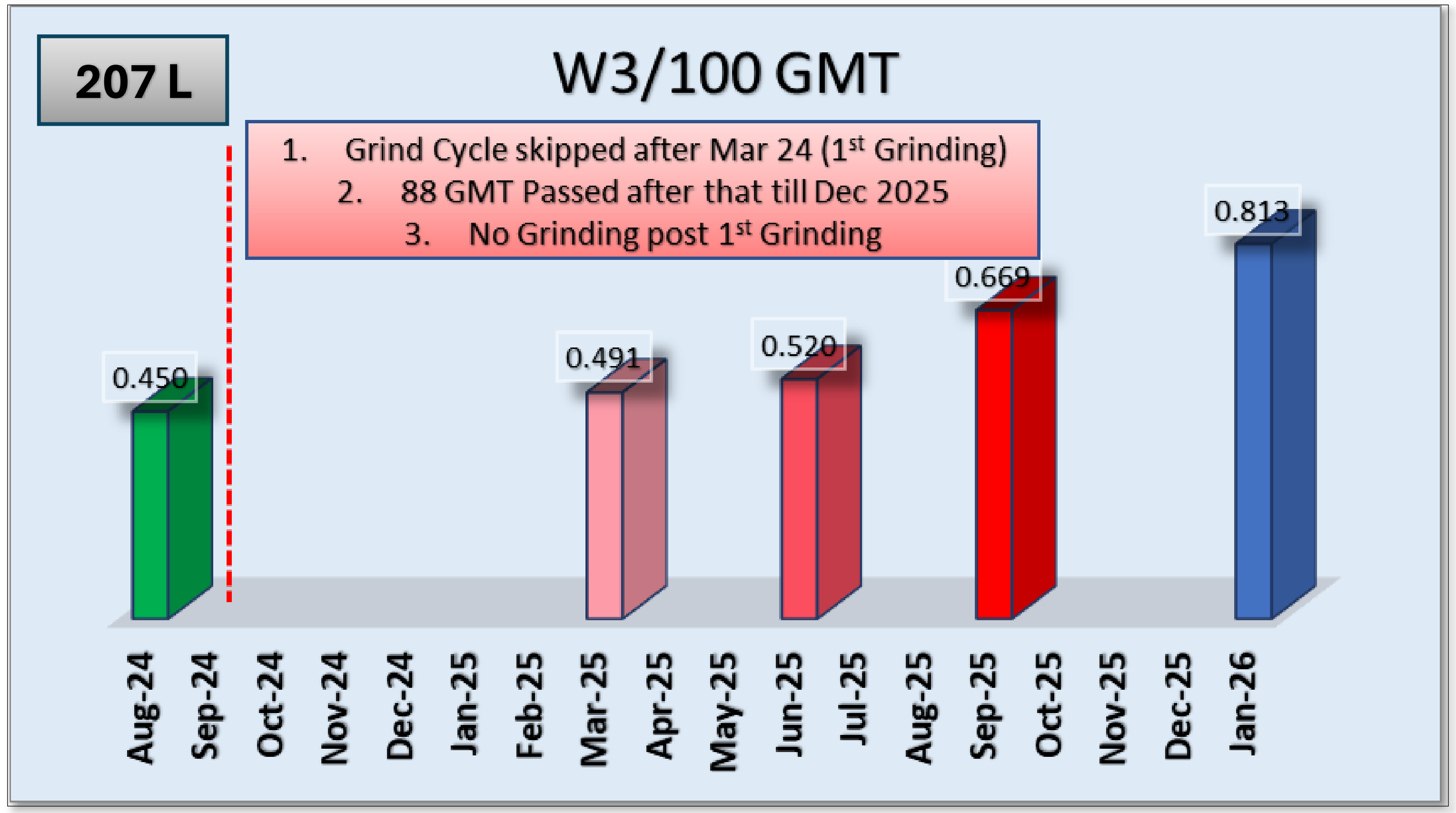

- For W1, the high rail wear rate increased from 0.186 in Aug-24 to 1.313 by Jan-26
- Similarly, for W3, the low rail wear rate rose from 0.450 in Aug-24 to 0.813 by Jan-26

Increase in wear rate (%) – Curve 207L

Test Site	Rail head angle	Mar 25 – Jun 25	Jun 25 – Sep 25	Sep 25 – Jan 26
207L - High	0 ˚ (W1)	67	73.3	66.08
207L - High	45 ˚ (W3)	6	28.7	21.50

- For the low rail, the wear rate increased progressively under W1 (0°) from 67% (Mar–Jun 2025) to 73.3% (Jun–Sep 2025) and 66.08% (Sep 2025–Jan 2026)
- while under W3 (45°) it increased from 6% to 28.7% and 21.5%

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Damage acceleration Behaviour Contd.

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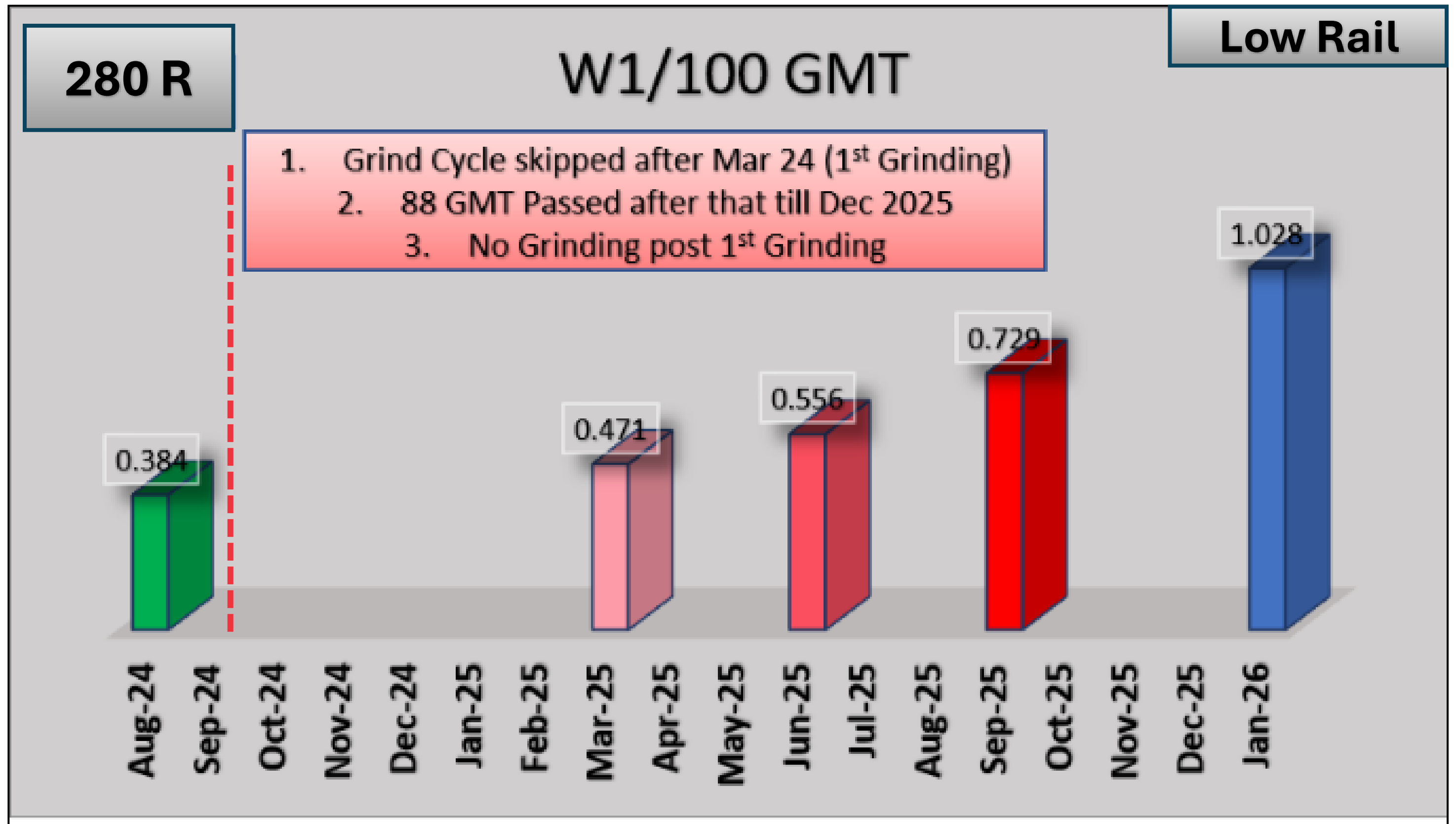

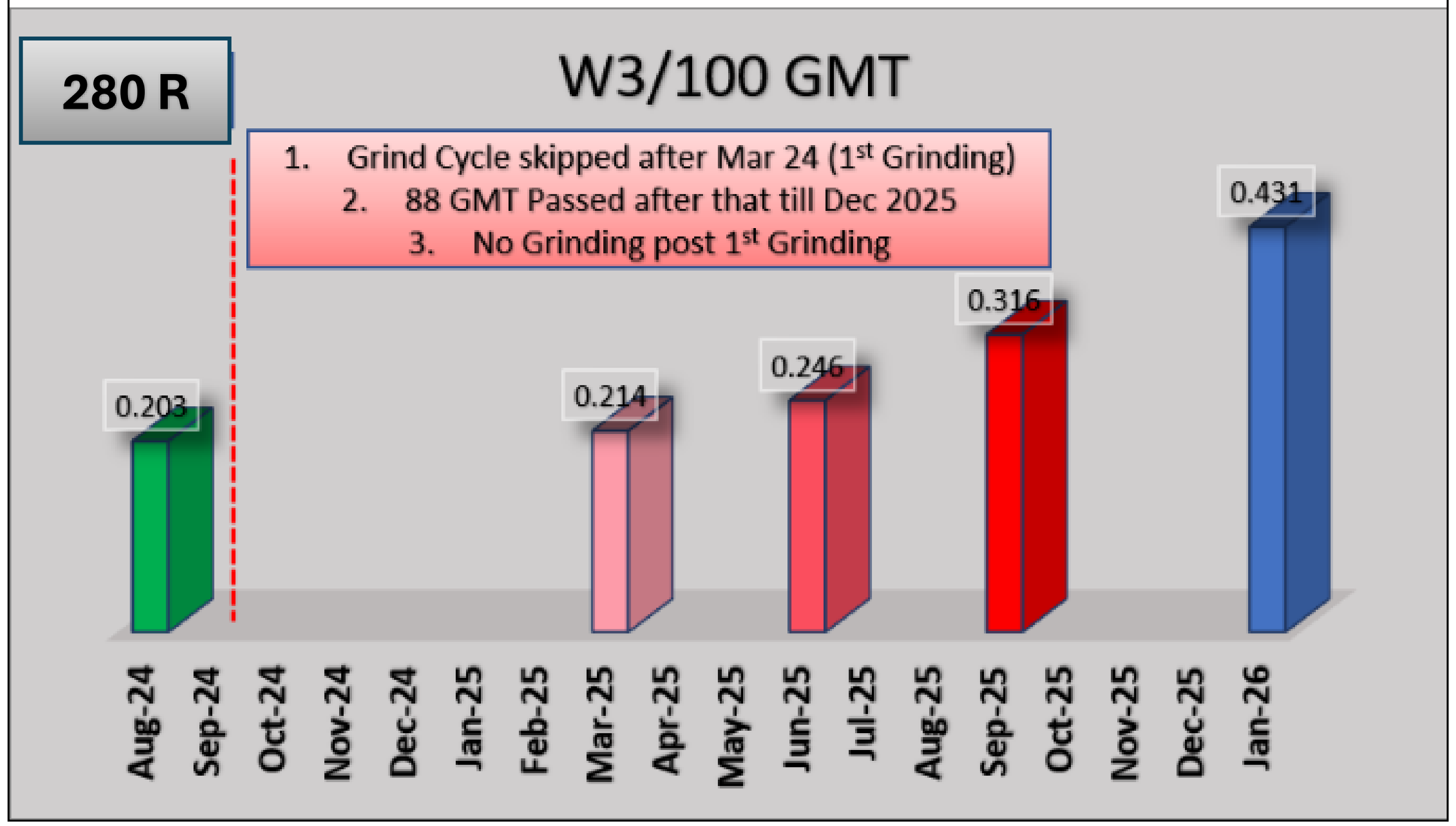

- The data indicate a consistent increase in wear rate with the accumulation of GMT.
- For W1, the low rail wear rate increased progressively from 0.384 in Aug-24 to 1.028 by Jan-26
- A similar trend is observed for W3, where the low rail wear rate increased from 0.203 in Aug-24 to 0.431 by Jan-26.

Increase in wear rate (%) – Curve 280R				
Test Site	Rail head angle	Mar 25 – Jun 25	Jun 25 – Sep 25	Sep 25 – Jan 26
280R - Low	0˚ (W1)	18.1%	31.15%	40.89%
280R - Low	45˚ (W3)	15.2%	28.4%	36.2%

- For the low rail, the wear rate increased progressively under W1 (0°) from 18.1% (Mar–Jun 2025) to 31.15% (Jun–Sep 2025) and 40.89% (Sep 2025–Jan 2026)
- while under W3 (45°) it increased from 15.2% to 28.4% and 36.2%

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Damage acceleration Behaviour Contd.

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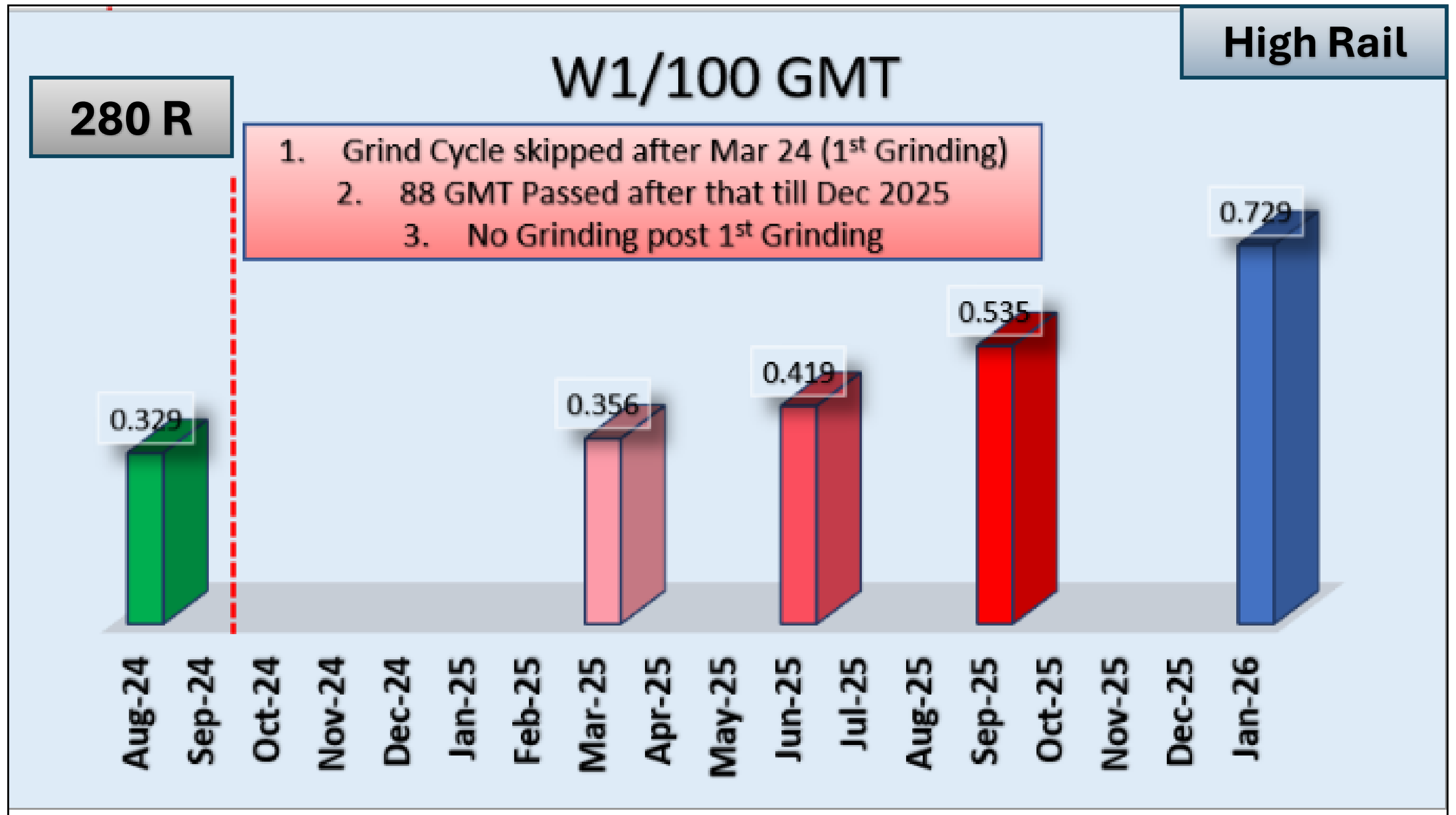

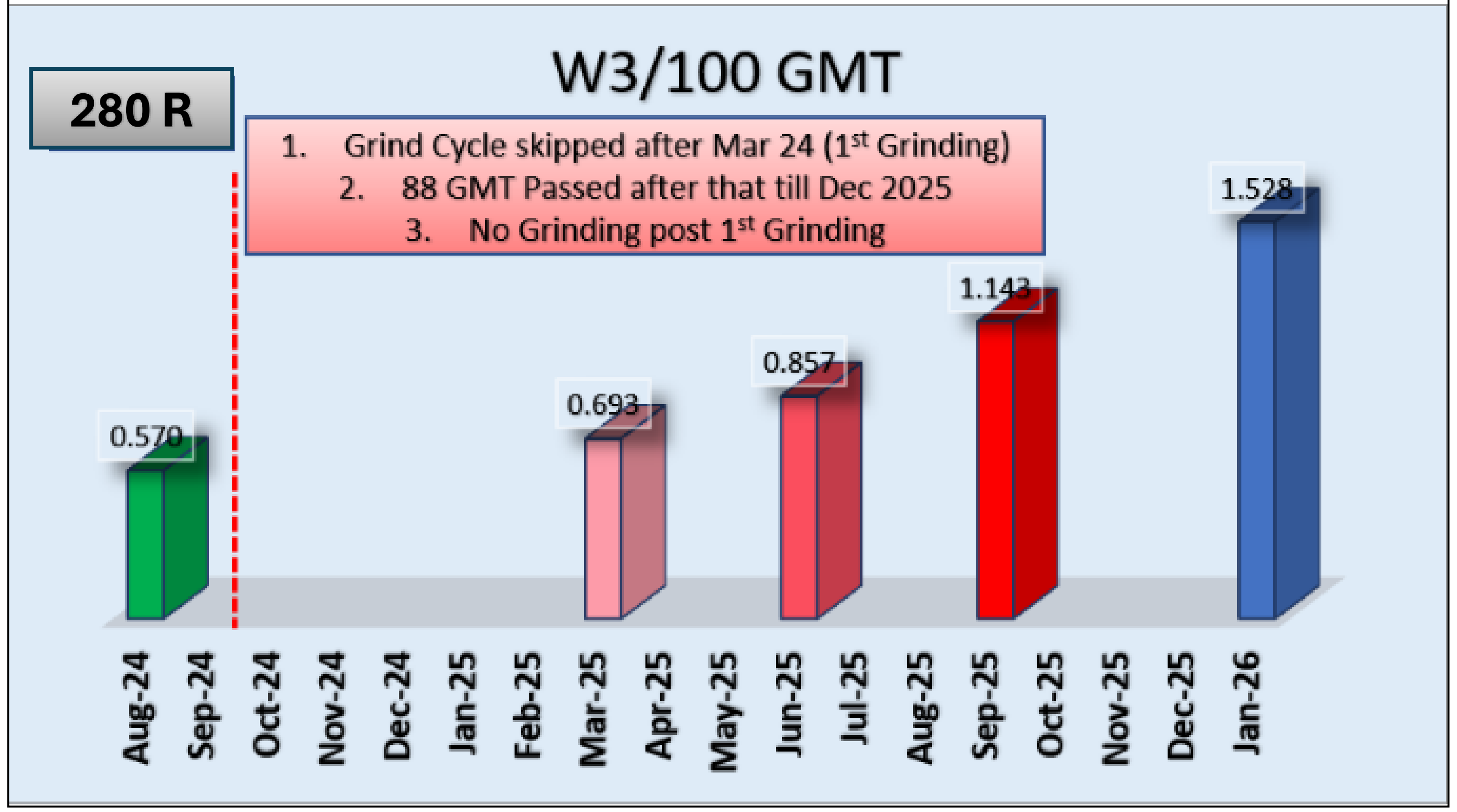

- For W1, the high rail wear rate increased from 0.329 in Aug-24 to 0.729 by Jan-26
- Similarly, for W3, the low rail wear rate rose from 0.570 in Aug-24 to 1.528 by Jan-26

Increase in wear rate (%) – Curve 280R

Test Site	Rail head angle	Mar 25 – Jun 25	Jun 25 – Sep 25	Sep 25 – Jan 26
280R - High	0˚ (W1)	17.8%	27.5%	36.3%
280R - High	45˚ (W3)	23.8%	33.3%	33.68%

- For the low rail, the wear rate increased progressively under W1 (0°) from 18.1% (Mar–Jun 2025) to 31.15% (Jun–Sep 2025) and 40.89% (Sep 2025–Jan 2026)
- while under W3 (45°) it increased from 15.2% to 28.4% and 36.2%

Dedicated Freight Corridor

Damage acceleration Behaviour Contd.

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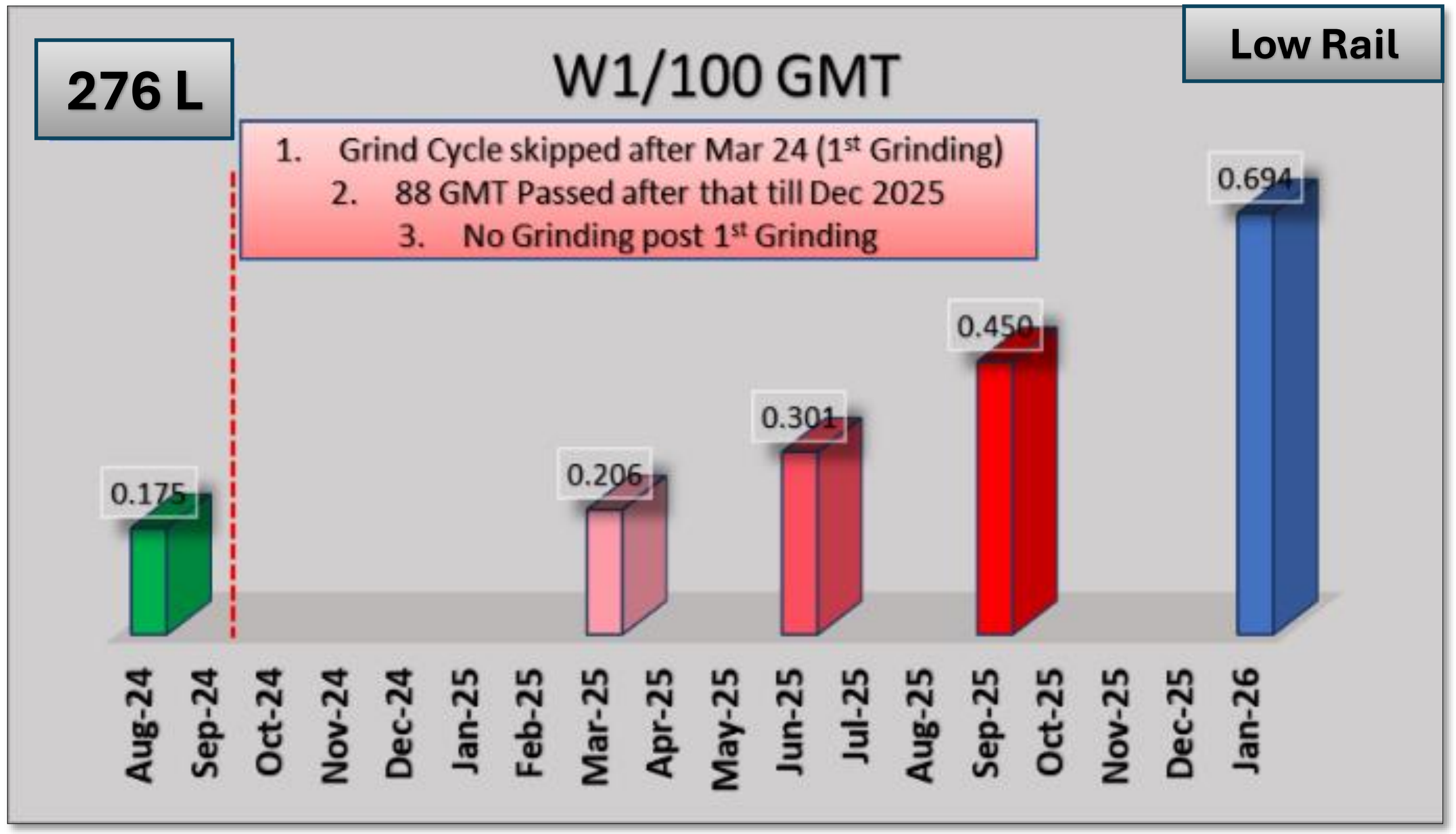

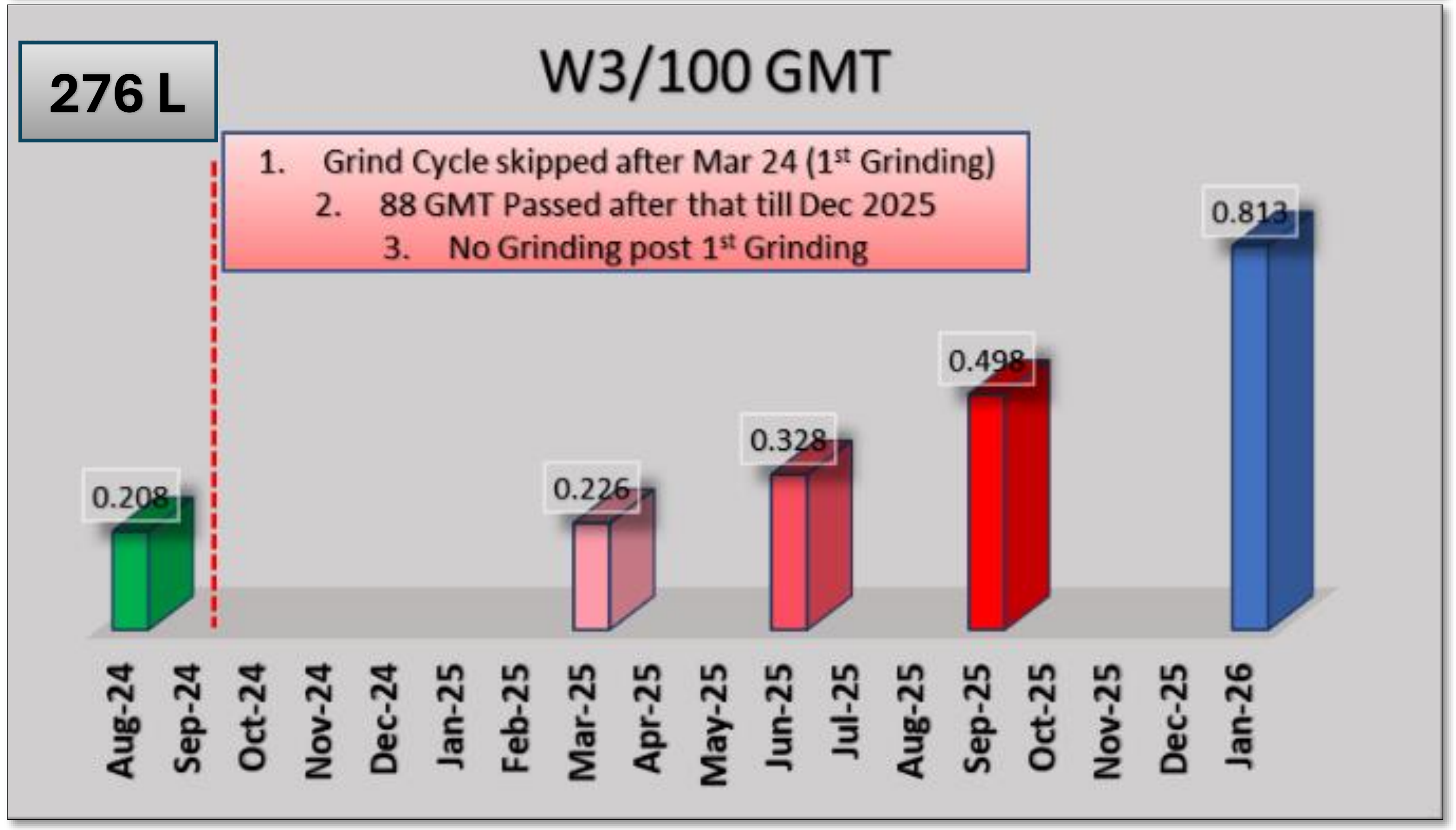

- For W1, the wear rate increased from 0.175 in Aug-24 to 0.694 by Jan-26, indicating a steady and continuous rise after the grinding cycle was deferred.
- Similarly, for W3, the wear rate increased from 0.208 in Aug-24 to 0.813 by Jan-26, showing a comparatively sharper escalation under angled contact conditions.

Increase in wear rate (%) – Curve 276L

Test Site	Rail head angle	Mar 25 – Jun 25	Jun 25 – Sep 25	Sep 25 – Jan 26
276L - Low	0˚ (W1)	46.2	49.5	54.7
276L - Low	45˚ (W3)	45.5	51.9	62.99

- For W1 (0°), the wear rate increased progressively from 46.2% (Mar–Jun 2025) to 49.5% (Jun–Sep 2025) and further to 54.7% (Sep 2025–Jan 2026), indicating sustained deterioration.
- For W3 (45°), the wear rate showed a more pronounced increase from 45.5% to 51.9% and further to 62.99%, highlighting accelerated wear progression under higher contact angles.

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Damage acceleration Behaviour Contd.

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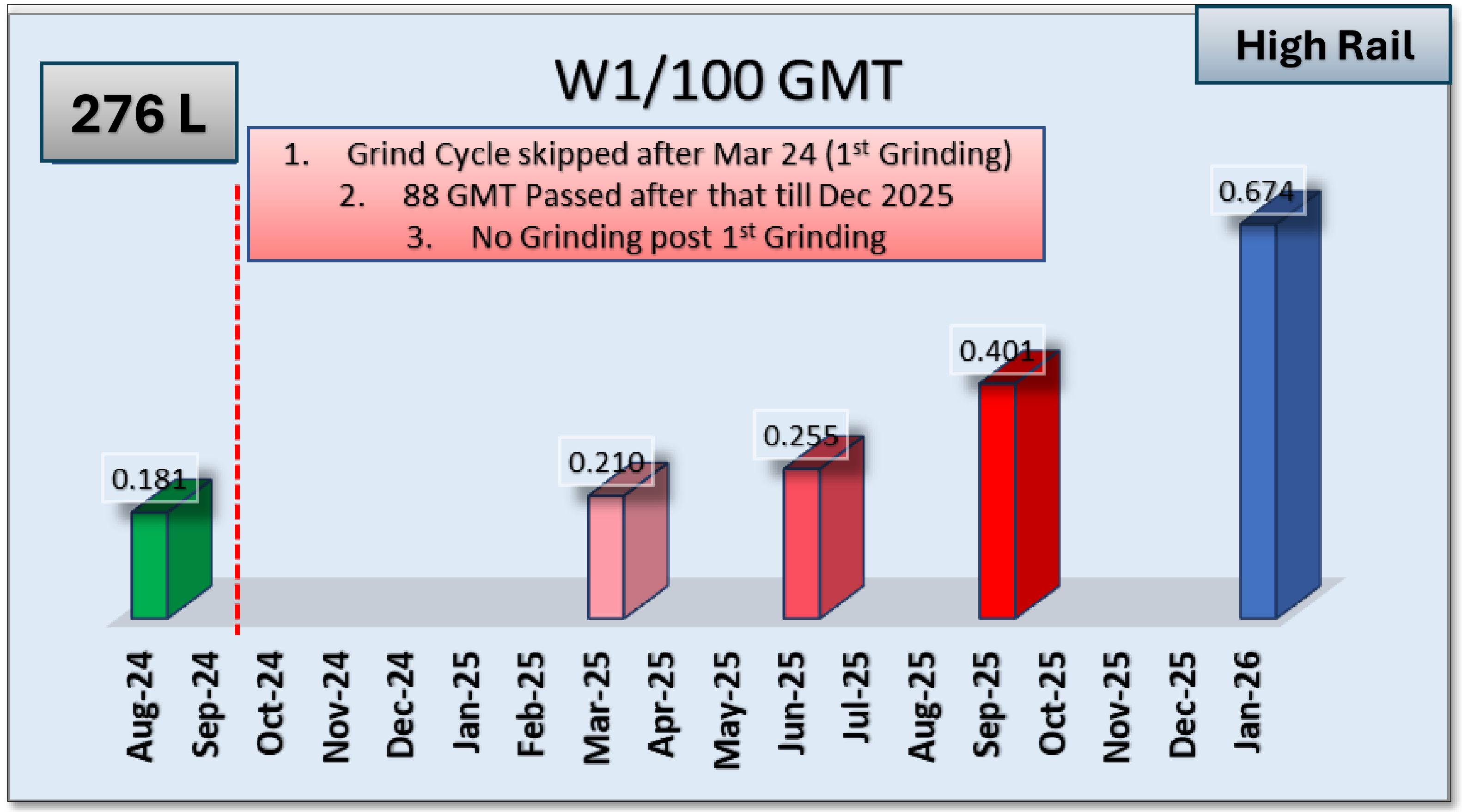

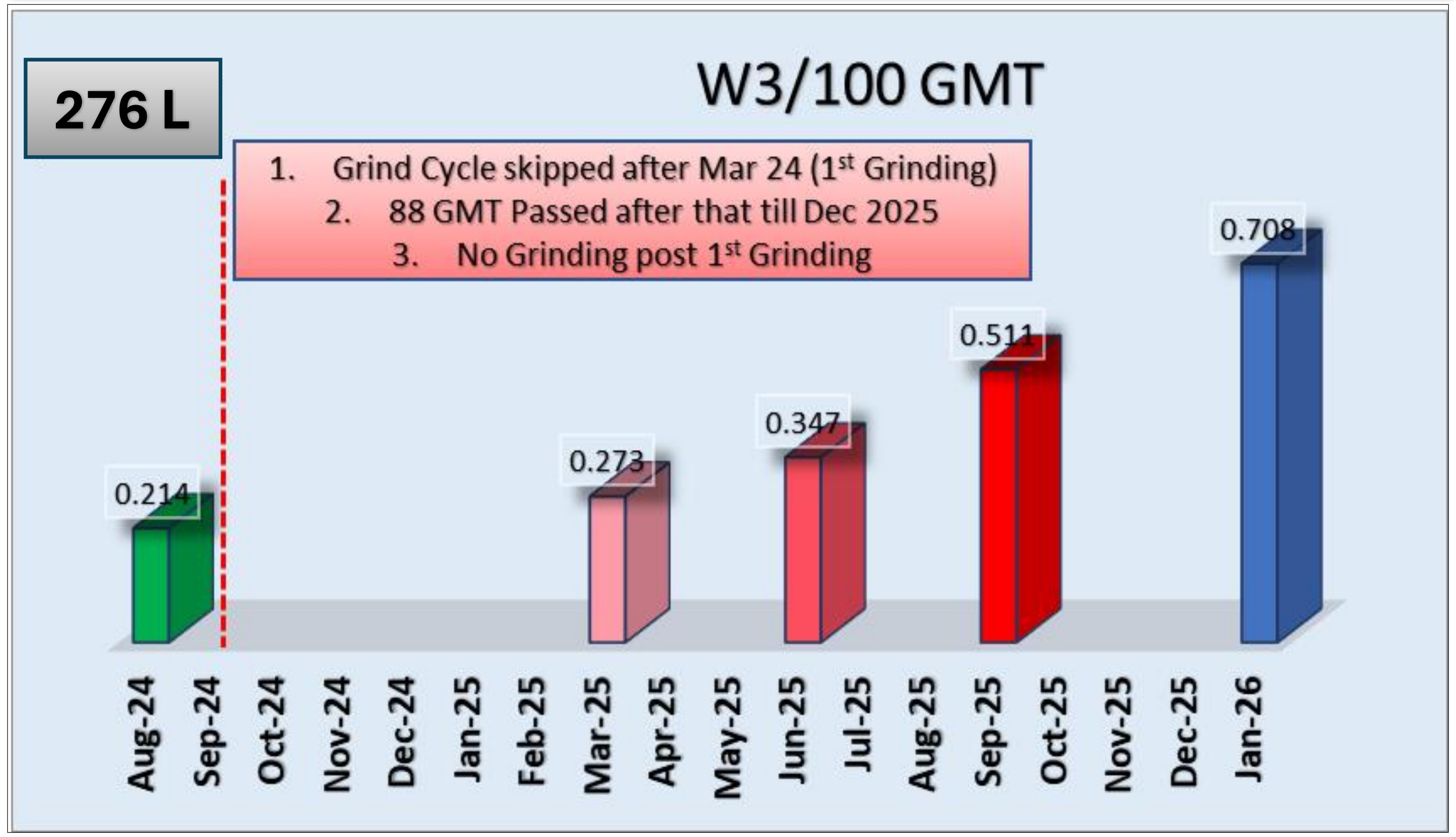

- For W1, the high rail wear rate increased from 0.181 in Aug-24 to 0.674 by Jan-26, showing a consistent rise following the skipped grinding cycle.
- Similarly, for W3, the wear rate increased from 0.214 in Aug-24 to 0.708 by Jan-26, indicating sustained deterioration under angled contact conditions.

Increase in wear rate (%) – Curve 276L

Test Site	Rail head angle	Mar 25 – Jun 25	Jun 25 – Sep 25	Sep 25 – Jan 26
276L – High	0˚ (W1)	21.7	57.1	67.89
267L - High	45˚ (W3)	26.91	47.37	38.71

- For W1 (0°), the wear rate increased sharply from 21.7% (Mar–Jun 2025) to 57.1% (Jun–Sep 2025) and further to 67.89% (Sep 2025-Jan 2026), indicating accelerated wear progression.
- For W3 (45°), the wear rate initially increased from 26.91% to 47.37%, but then reduced to 38.71% in the final period, suggesting a relative stabilization compared to W1.

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Damage acceleration Behaviour (Inflection Point)

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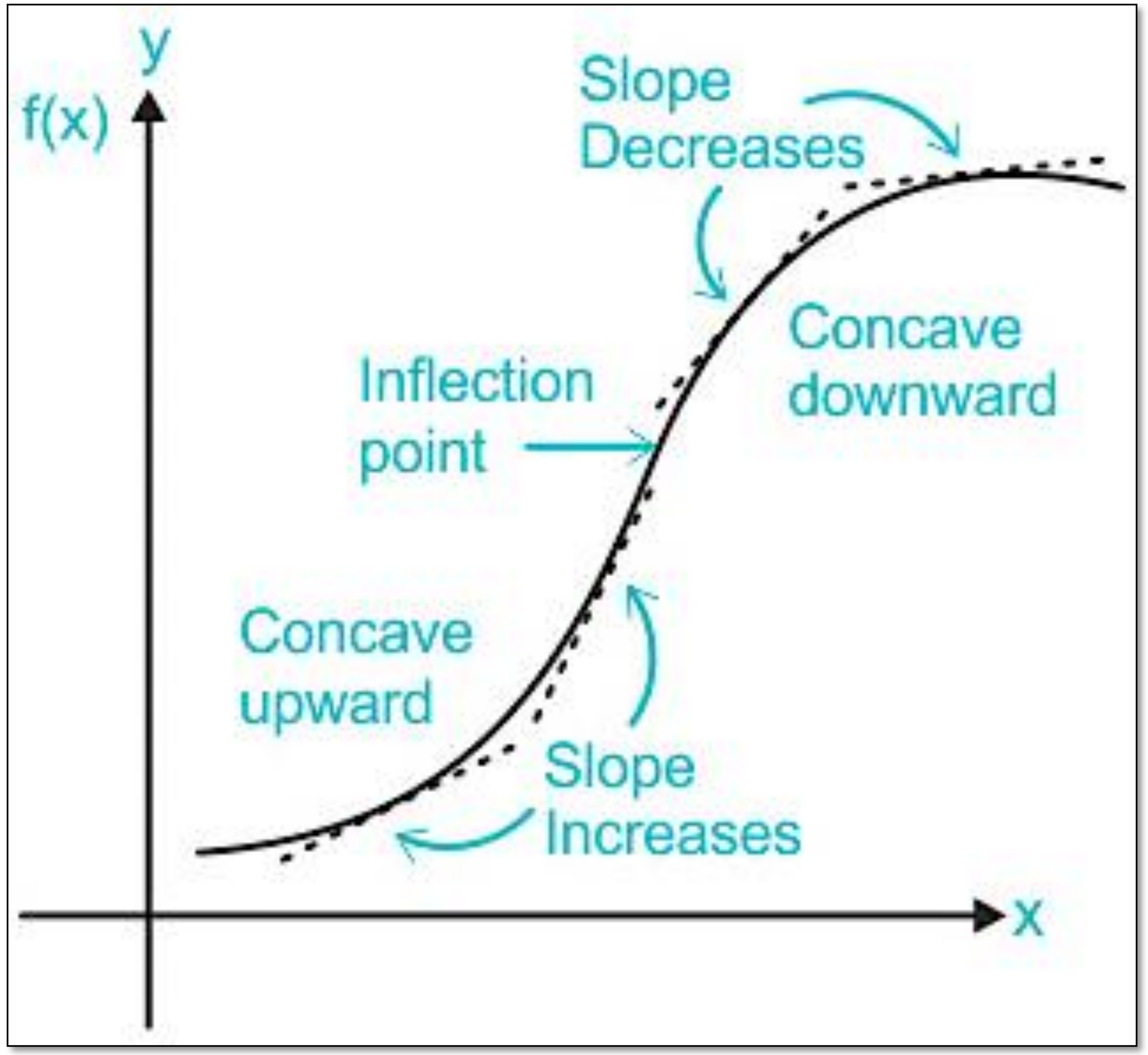


Inflection point is the point on a curve where the rate of increase/decrease changes significantly, indicating a shift in behavior.

Data Basis

Wear rates (mm/100 GMT) were obtained from periodic measurements across:

- Low Rail and High Rail
- W1 (0°) and W3 (45°) contact conditions
- Data points span increasing GMT intervals, capturing progressive rail degradation

Methodology Adopted

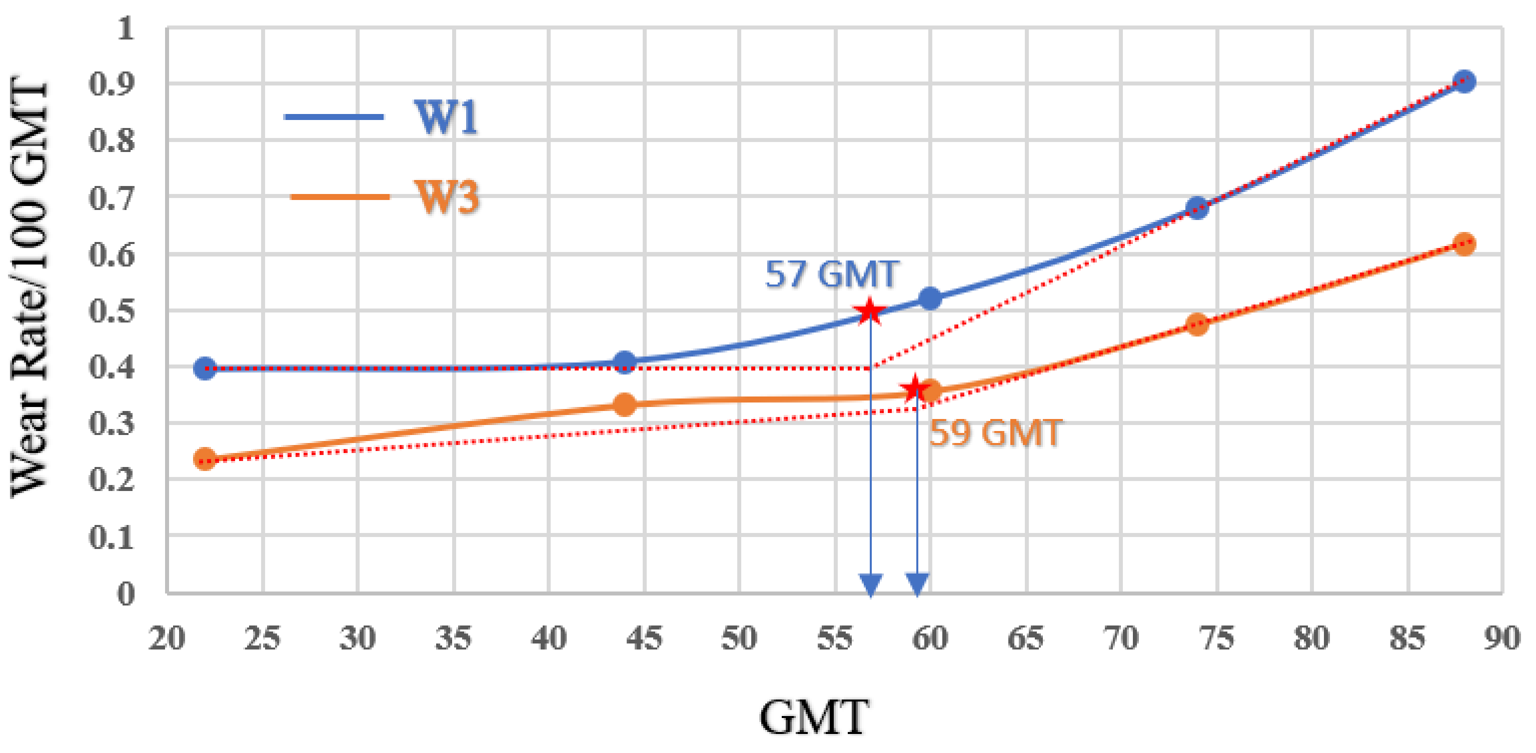
- Wear rate vs GMT plots were generated for each rail condition (W1 & W3)
- The trend was analyzed for non-linearity, identifying the point where slope changes from mild (linear/gradual) to steep (accelerated increase)
- The inflection point was determined as the GMT corresponding to the onset of sharp increase in wear rate (change in slope)

Engineering Significance

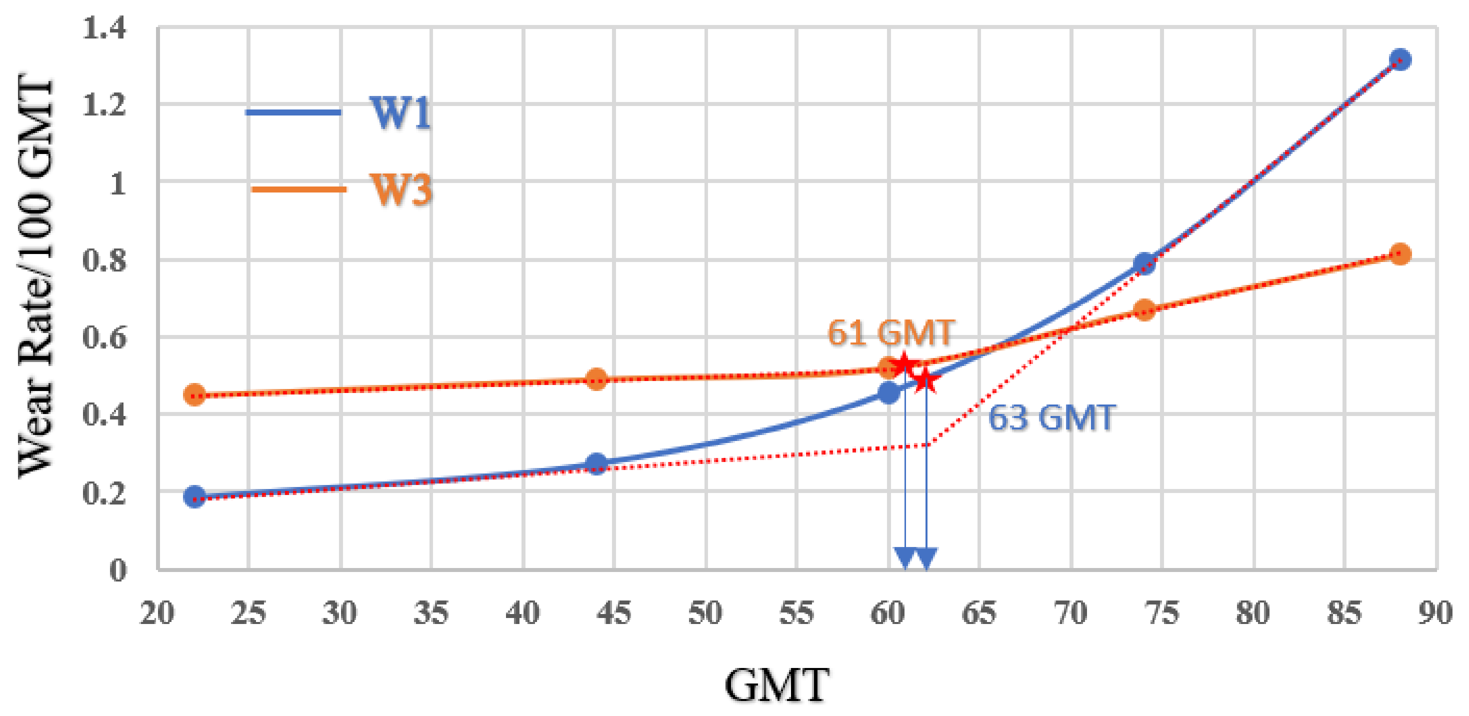
"Defines the upper limit for preventive grinding intervention"

Damage acceleration Behaviour Contd.

Inflection Point – Low Rail



Inflection Point – High Rail



Sharp - 207L – 2.5˚

Rail Position	Wear Parameter	Inflection Point (GMT)
Low Rail	W1 (0°)	57 GMT
Low Rail	W3 (45°)	59 GMT
High Rail	W1 (0°)	63 GMT
High Rail	W3 (45°)	61 GMT
Average Inflection Point		***60 GMT**

- The identified inflection points occur at around 59 GMT for W3 and 57 GMT for W1 on the low rail
- For the high rail, a similar nonlinear trend is observed, with inflection points occurring at approximately 61 GMT for W3 and 63 GMT for W1
- Beyond the inflection region, both W1 and W3 exhibit a steeper increase, confirming the onset of accelerated wear and fatigue-dominated behaviour.
- The average inflection point found out to be **60 GMT**

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Damage acceleration Behaviour Contd.

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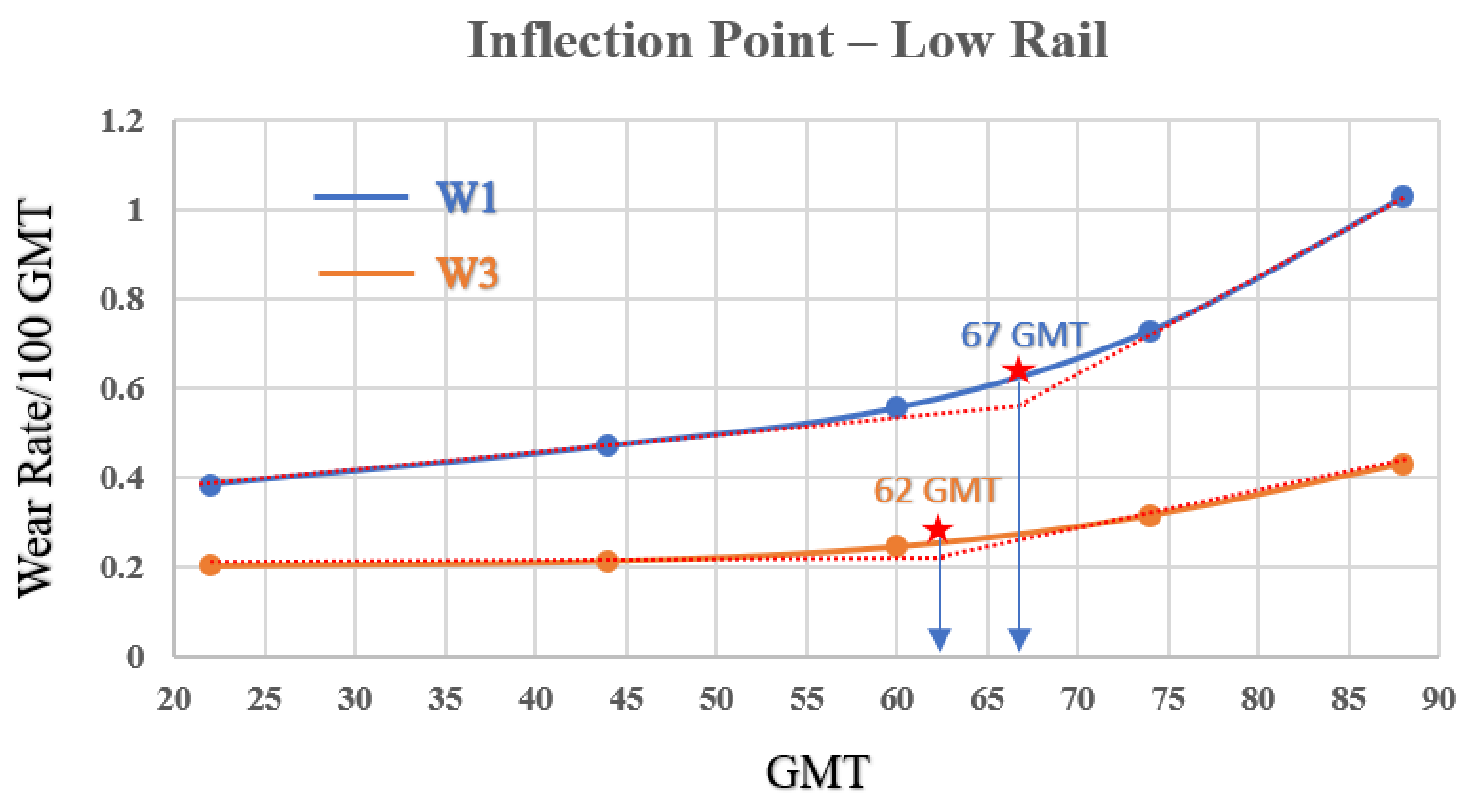
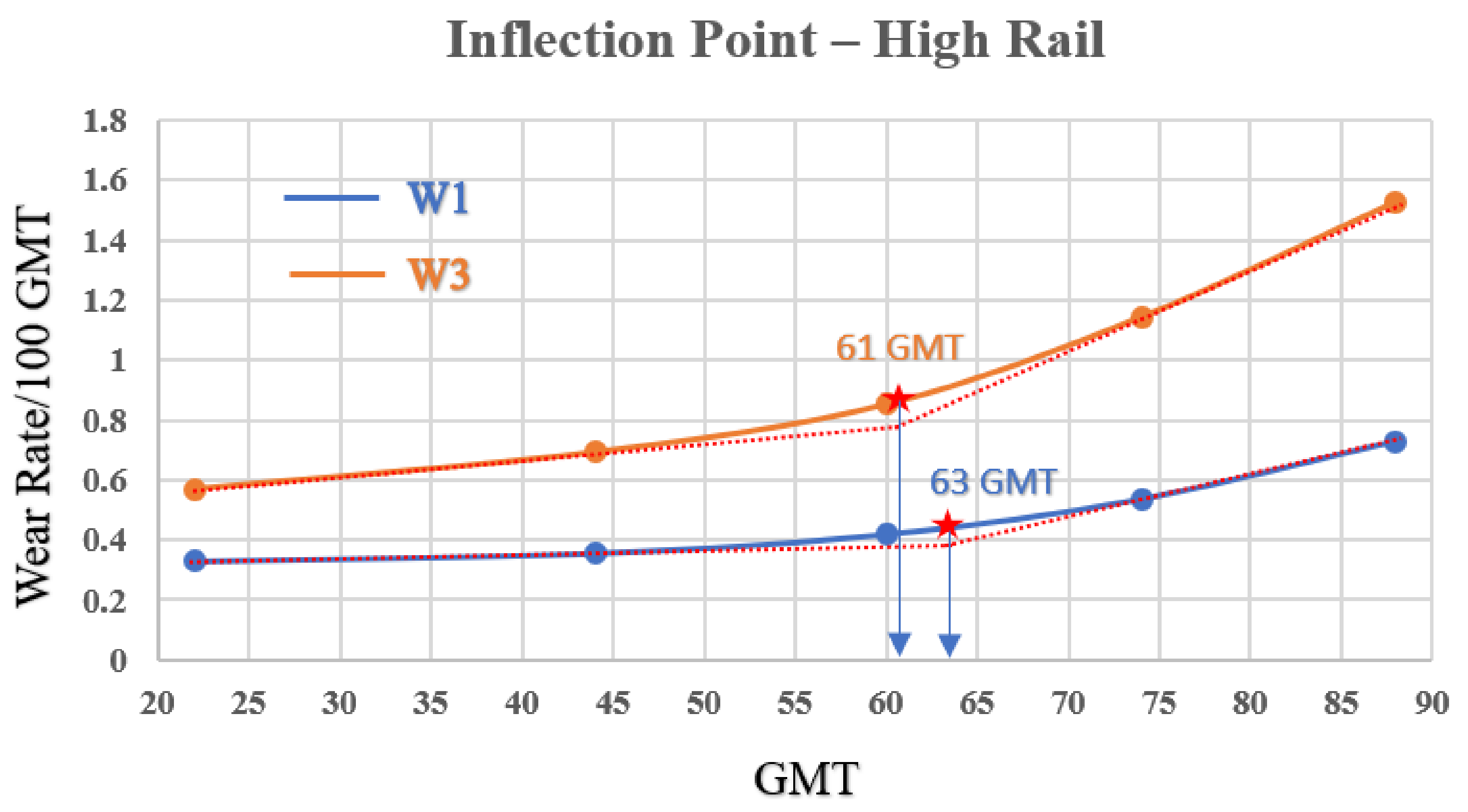
Sharp – 280R – 2.5˚





Rail Position	Wear Parameter	Inflection Point (GMT)
Low Rail	W1 (0°)	67 GMT
Low Rail	W3 (45°)	62 GMT
High Rail	W1 (0°)	63 GMT
High Rail	W3 (45°)	61 GMT
Average Inflection Point		***63.25 GMT**

- The identified inflection points occur at around 62 GMT for W3 and 67 GMT for W1 on the low rail
- For the high rail, a similar nonlinear trend is observed, with inflection points occurring at approximately 61 GMT for W3 and 63 GMT for W1
- Beyond the inflection region, both W1 and W3 exhibit a steeper increase, confirming the onset of accelerated wear and fatigue-dominated behaviour.
- The average inflection point found out to be **63.25 GMT**

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Damage acceleration Behaviour Contd.

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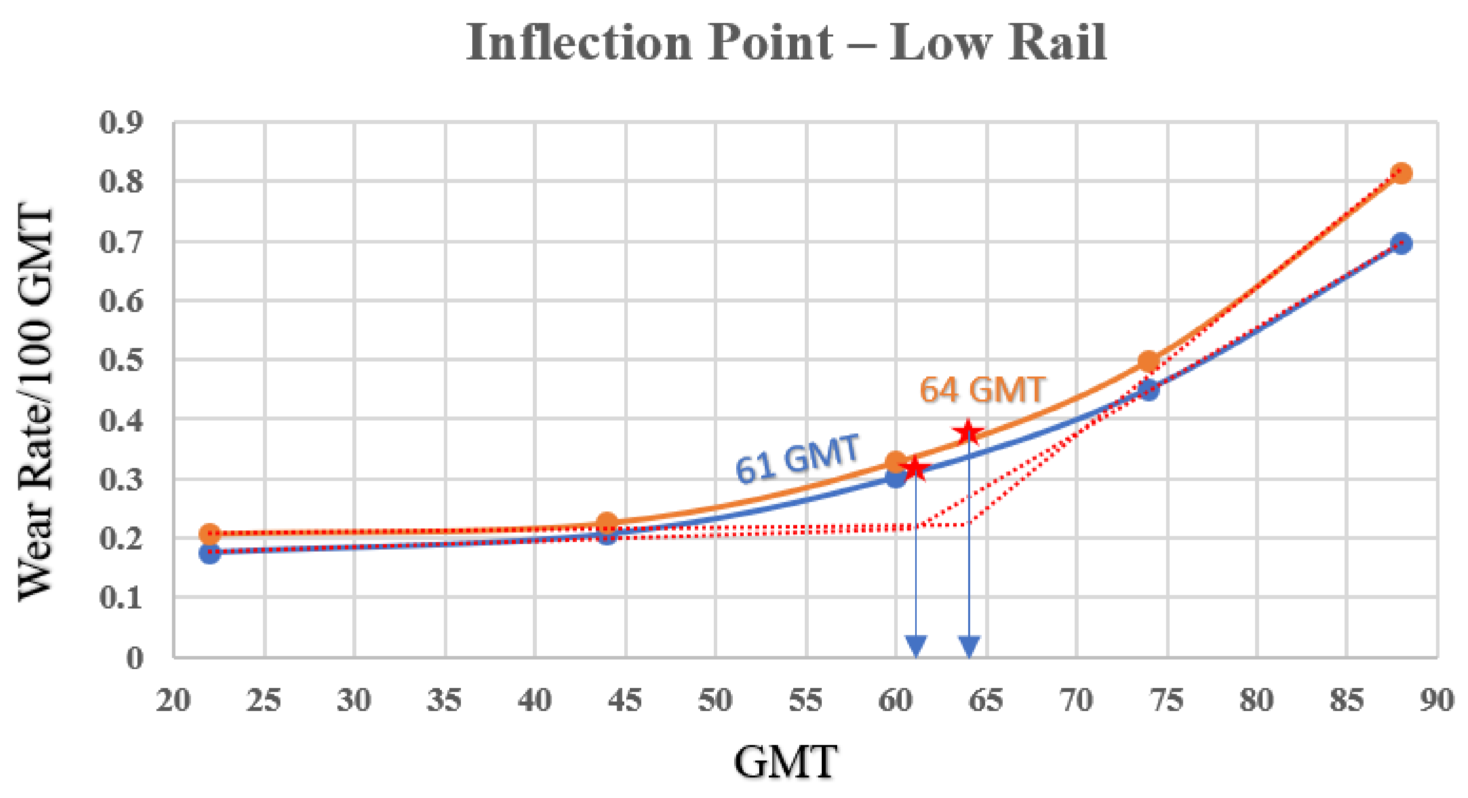
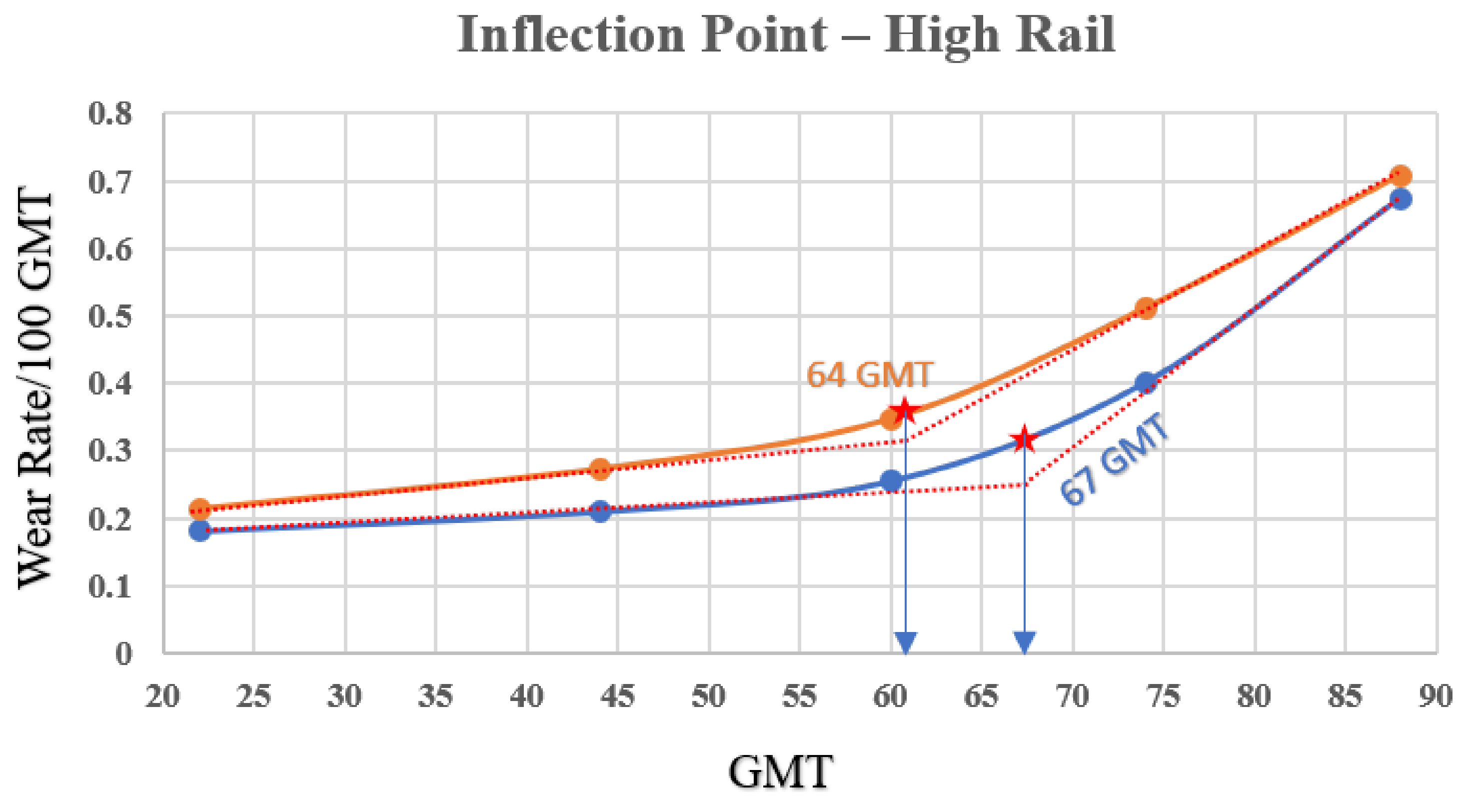
Sharp – 276L – 2.5˚





Rail Position	Wear Parameter	Inflection Point (GMT)
Low Rail	W1 (0°)	61 GMT
Low Rail	W3 (45°)	64 GMT
High Rail	W1 (0°)	67 GMT
High Rail	W3 (45°)	64 GMT
Average Inflection Point		***64 GMT**

- The identified inflection points occur at around 64 GMT for W3 and 61 GMT for W1 on the low rail
- For the high rail, a similar nonlinear trend is observed, with inflection points occurring at approximately 64 GMT for W3 and 67 GMT for W1
- Beyond the inflection region, both W1 and W3 exhibit a steeper increase, confirming the onset of accelerated wear and fatigue-dominated behaviour.
- The average inflection point found out to be **64 GMT**

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Material Degradation

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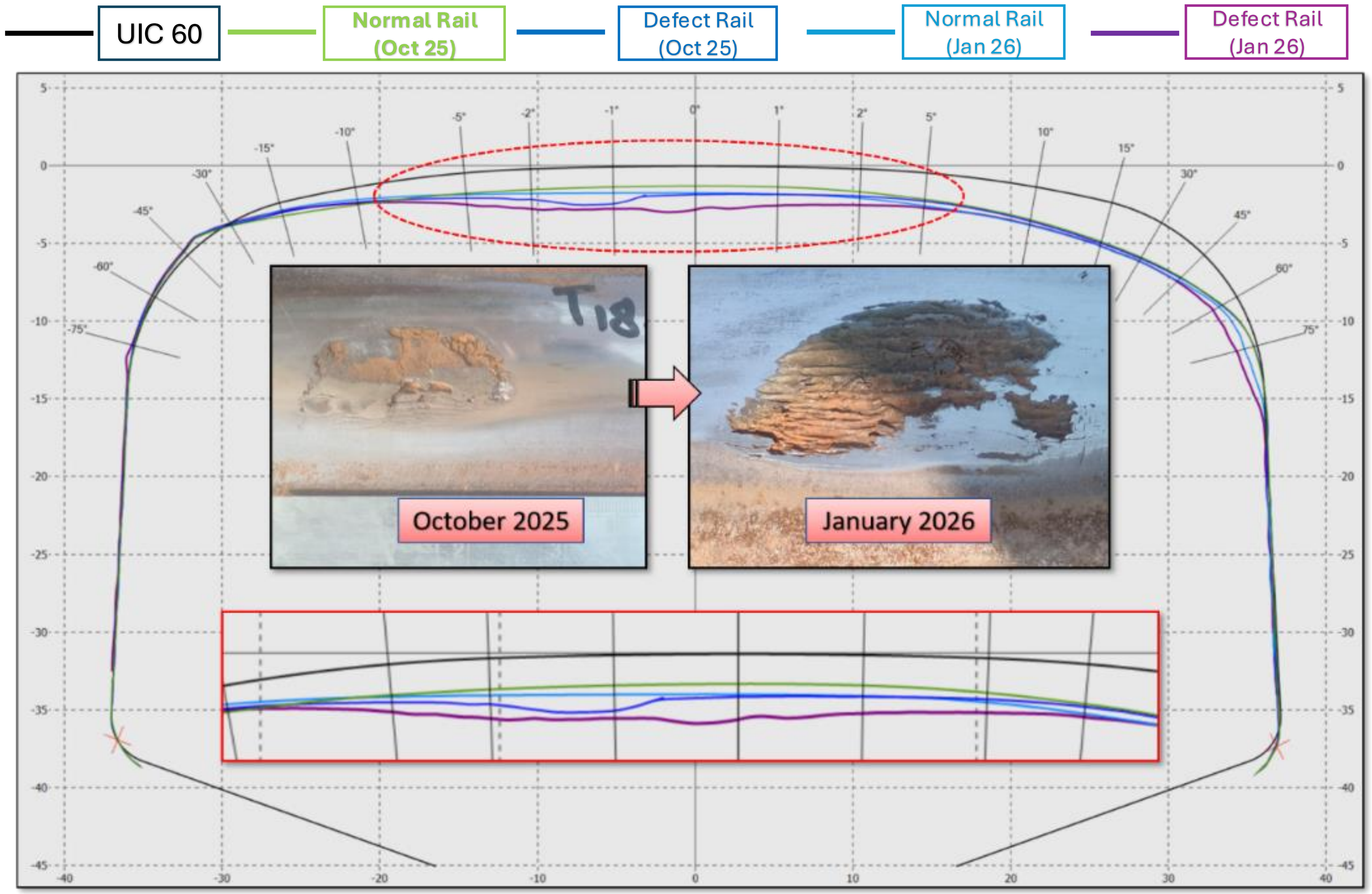

- The damage is limited to a localized area where the surface material has started to break away due to rolling contact fatigue.
- The rail profile plots also indicate the loss of metal on the rail head.
- The comparison clearly demonstrates the progressive evolution of surface defect over a short period.
- This indicates that rail degradation is not only geometric in nature but also involves changes in both mechanical and metallurgical properties

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Material Degradation Contd.

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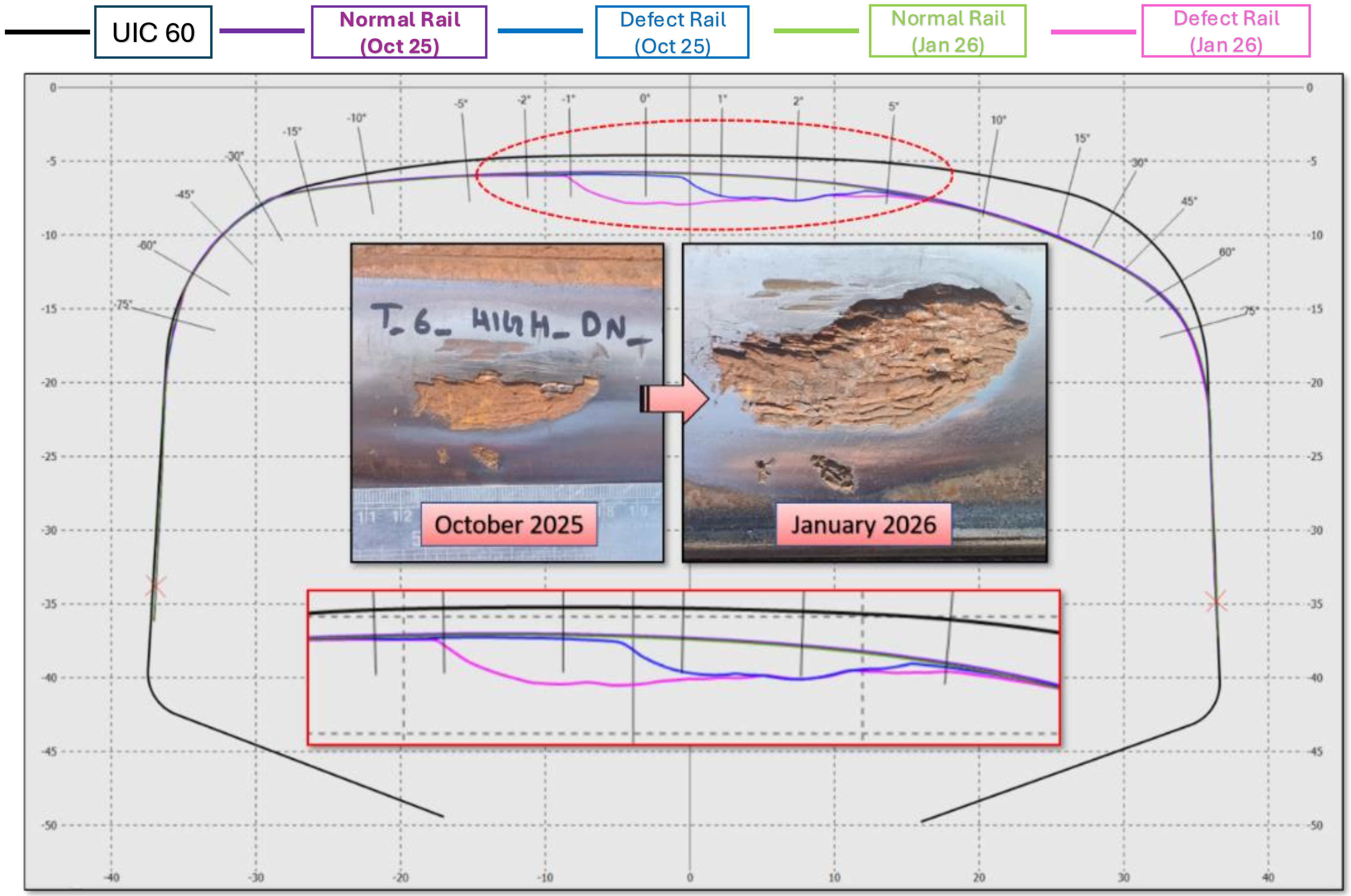

- The damage is limited to a localized area where localized material separation was observed due to rolling contact fatigue.
- The rail profile plots also indicate the loss of metal on the rail head.
- The comparison clearly demonstrates the rapid progression of the defect over a short period.
- This indicates that rail degradation is not only geometric in nature but also involves changes in both mechanical and metallurgical properties

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Sensitivity of Head hardened rails to Grinding Deferral

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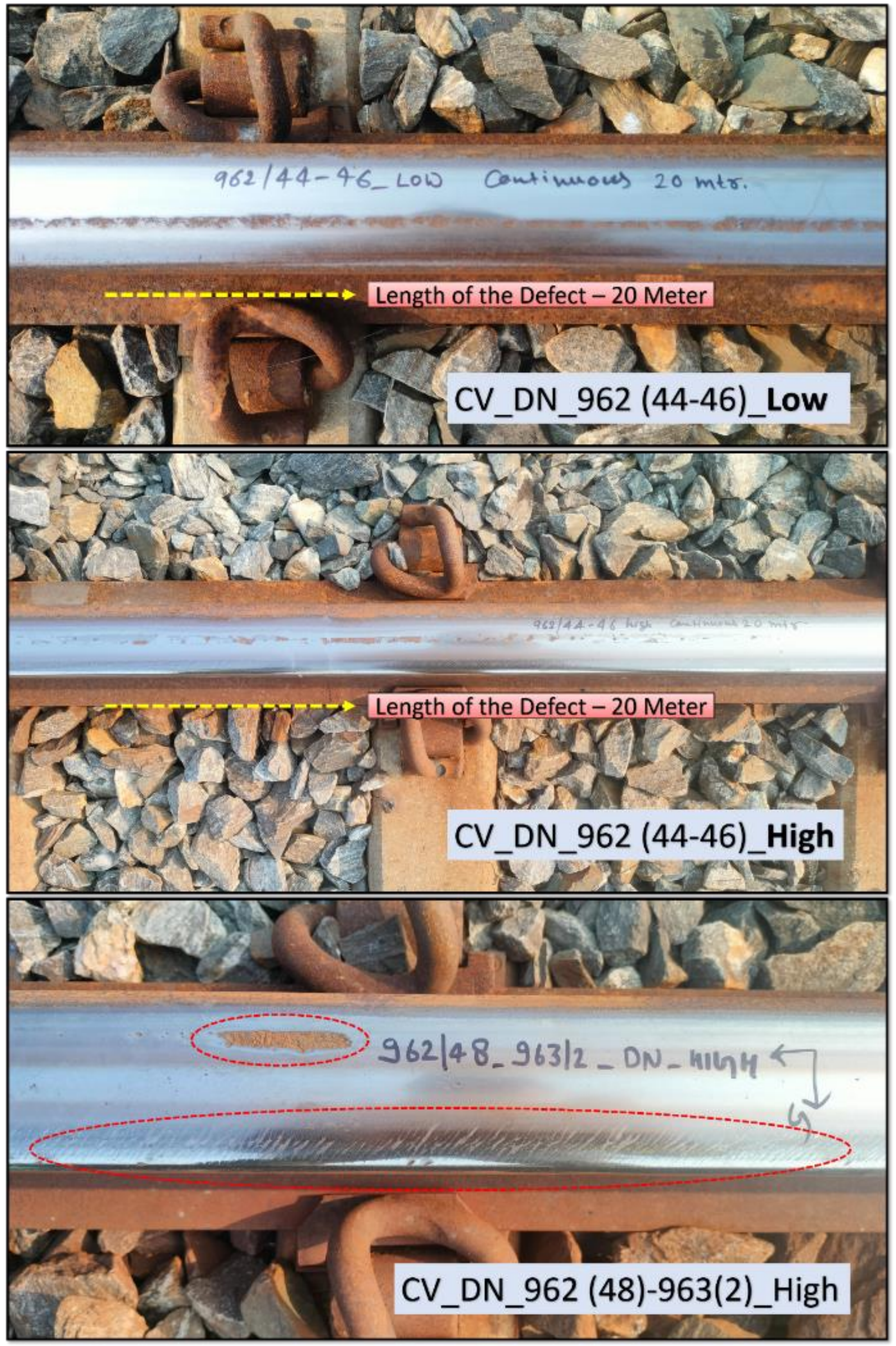

- Material flaking and continuous running band defects are observed on both high and low rails
- Rail surfaces exhibit elongated and irregular patches where the top metal layer has detached along the running band
- In several locations, these features observed over extended track lengths, forming continuous surface response patterns along the contact band
- Despite their superior wear resistance and higher hardness, 1080 head hardened (HH) rails exhibit increased sensitivity to deferred grinding. Preventive grinding seems an essential task to be carried out in a planned manner.

Rail Grinding as a Reset Mechanism

"Continuous or frequent removal of a small amount of surface material truncate existing cracks and can theoretically eliminate surface crack initiation altogether".

1970 - In Japan, the operation has been introduced to the Tokaido Shinkansen line
Early 1980's - Canadian Pacific Railway (CPR) was using profile grinding as a means of controlling rail problems
Since the 1980s, rail grinding technology has been applied to the rail maintenance

Author	Statement	Year
Ishida	Reported good results with **rail grinding** of squats in Japan	2003
Jonas Lundmark	One of the important aspects in wheel/rail interface management is the introduction of **rail grinding**	2007
Yukio Satoh	**Rail grinding** operation has been introduced in order to extend the service life of rail by removing the RCF layer	2008
Roger Singleton	**Rail grinding** has been identified by Network Rail as a key maintenance activity	2015
M. Mesaritis	**Rail grinding** is used to minimize these external; and internal defects and reshape the rail line	2020

Rail Grinding Operation

Inferences - WDFC

- Significant increase in cumulative GMT observed across the WDFC network year-on-year, reflecting rising traffic intensity and an associated escalation in rail degradation rates.
- With new sections added to connect the ports, the traffic is expected to rise even further.
- Continued reliance on corrective grinding resulting in localized interventions, leading to partial track coverage and resulting in prioritized coverage of critical sections only, leaving substantial stretches of the WDFC network unaddressed.
- Opportunity to transition towards a data driven, preventive rail grinding program.
- ***A consistent inflection point in the degradation curve marks the onset of accelerated damage and can be used as a framework for determining grinding intervals instead of fixed GMT bands.***
- *It should be noted that wear alone cannot determine grinding frequency. We should also take into account RCF, surface defects and Defect Generation Rates (DGR) from ultrasonic testing.*

Rehabilitation measures taken:

Based on the study outcomes, DFCCIL has proactively strengthened its rail maintenance strategy by deploying one SRGM (20 stone) machine, with plans to induct a 96-stone grinding unit to support increasing GMT and achieve network-wide grinding in accordance with prescribed maintenance cycles.

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References

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1. Dou, S., Lin, D., Huang, X., Wang, W., Ding, H., Zhang, S., Yang, W., & Zhou, Z. (n.d.). *Effect of material characteristics and axle loads on heavy-haul railway train wheel wear and rolling contact fatigue behavior*. Retrieved https://ssrn.com/abstract=5347505
2. Evans, G. (2013). Managing rail profile. *Ironmaking and Steelmaking*, *40*(2), 115–119. https://doi.org/10.1179/1743281212Y.0000000033
3. Magel, E. E., & Kalousek, J. (2002). The application of contact mechanics to rail profile design and rail grinding. In *Wear* (Vol. 253).
4. Singleton, R., Marshall, M. B., Lewis, R., & Evans, G. (2015). Rail grinding for the 21st century - Taking a lead from the aerospace industry. In *Proceedings of the Institution of Mechanical Engineers, Part F: Journal of Rail and Rapid Transit* (Vol. 229, Number 5, pp. 457–465). SAGE Publications Ltd. https://doi.org/10.1177/0954409714527929
5. Steenbergen, M. (2016). Rolling contact fatigue in relation to rail grinding. *Wear*, *356–357*, 110–121. https://doi.org/10.1016/j.wear.2016.03.015

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